TOPIC 2 · FLOW KINEMATICS

Solved Problems: Official Collection K1 to K10

Worked solutions to the ten problems of the flow-kinematics collection of the Department of Thermal and Fluid Engineering, Universidad Carlos III de Madrid. Every problem restates the official statement, then follows four phases: (1) hypotheses, degrees of freedom and data; (2) coordinates, frames and changes of variable; (3) step-by-step derivation; (4) interpretation, limits and dimensional check. Discrepancies found in the handwritten official solutions are flagged in the corresponding problem. References to Notes.pdf are to Chapter 2 (Flow kinematics).

PROBLEM K1

Stokes layer · Path lines · Fluid line · Circulation

Flow over an Oscillating Porous Wall with Transverse Blowing

Original statement · K1.pdf

OFFICIAL PROBLEM STATEMENT

A fluid is blown with velocity V through a horizontal porous wall that is oscillating in its plane with velocity $U\cos(\Omega t)$. The resulting fluid velocity for $0 < y < \infty$ is given by (approximately)

$$v_x = U e^{-y/\delta} \cos\left(\Omega t - \frac{y}{\delta}\right), \qquad v_y = V,$$

where the characteristic thickness $\delta = (2\nu/\Omega)^{1/2}$ depends on the kinematic viscosity of the fluid ν.

1. Determine the trajectory and path of a fluid particle initially located at $x = 0$ and $y = 0$.
2. Obtain the streamline that intersects the point $x = 0$ and $y = 0$.
3. Compute the vorticity $\nabla \wedge \vec{v}$. If $\nabla \wedge \vec{v} = 0$, obtain also the velocity potential.
4. Calculate the rate of expansion $\nabla \cdot \vec{v}$. If $\nabla \cdot \vec{v} = 0$, obtain also the stream function.
5. Equation of the fluid line formed by the fluid particles initially located along the vertical axis $x = 0$.
6. Determine the volume flux crossing the plane $x = 0$ as a function of time $Q(t)$ along with its average value $Q_a = \int_t^{t+2\pi/\Omega} Q\, dt$.
7. Obtain the volume flux leaving the sphere of radius R centered at $x = 0$ and $y = 2R$.
8. Compute the circulation around a square of side L whose base lies parallel to the wall at a distance $y = L/2$. Redo the computation making use of Stokes theorem.

> Source: K1.pdf (page 1). Notes.pdf, Chapter 2: trajectories and path lines Eqs. (2.10)-(2.15), fluid lines Eqs. (2.16)-(2.17), stream lines Eq. (2.21), circulation and Stokes theorem Eqs. (2.29)-(2.33), potential Eq. (2.34), convective flux Eqs. (2.35)-(2.38), stream function Eqs. (2.40)-(2.45).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- The motion is planar: the velocity has components (v_x, v_y) that depend only on y and t. The only spatial degree of freedom of the field is y; the flow is unsteady with the single time scale Ω^{-1}.
- The field is given as an approximation to a viscous (Stokes-layer) solution; here it is treated as a prescribed kinematic field and no dynamics are used.
- The fluid occupies $0 < y < \infty$. At the wall, $v_x(y=0) = U\cos\Omega t$ equals the wall velocity and $v_y = V$ is the blowing velocity.
- We assume $V \neq 0$ (transverse blowing) and $\Omega > 0$, so that δ is real and positive.
- Fluxes and the circulation are computed per unit length in the direction z normal to the plane of the motion.

Symbol	Meaning	SI unit
U	amplitude of the wall velocity	m/s
V	blowing velocity (uniform, along y)	m/s
Ω	angular frequency of the wall oscillation	s^{-1}
ν	kinematic viscosity	m^2/s
$\delta = (2\nu/\Omega)^{1/2}$	thickness of the oscillatory layer	m
L, R	side of the square and radius of the sphere	m
$Q(t)$	volume flux per unit depth through $x=0$	m^2/s
Γ	circulation	m^2/s

Dimensional check of δ: $[\nu/\Omega] = (\mathrm{m}^2/\mathrm{s})/(1/\mathrm{s}) = \mathrm{m}^2$, so δ is a length.

PHASE 2

Coordinates, Frames and Changes of Variable

- Frame of observation: inertial laboratory frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, with x along the wall, y normal to it and $\mathbf{e}_z = \mathbf{e}_x \wedge \mathbf{e}_y$. The wall oscillates in this frame; the velocity field given in the statement is the laboratory-frame field.
- Eulerian variables: (x, y, t). Lagrangian label of a particle: its position (x_0, y_0) at $t = 0$.
- Phase variable used throughout: $\Theta(y,t) \equiv \Omega t - y/\delta$, so that $v_x = U e^{-y/\delta} \cos\Theta$.
- Change of variable along a particle: since $v_y = V$ is constant, $y(t) = y_0 + Vt$, hence $t = (y - y_0)/V$. This trades time for y when the path line is required.
- Abbreviations: $b \equiv \Omega - V/\delta$ and $D \equiv (V/\delta)^2 + b^2$ (so $D > 0$ because $V \neq 0$).

Why a lemma is needed

Parts 1, 2, 5 and 6 require primitives of an exponential times a cosine. Instead of quoting a table, the primitive is proved once. For constants a, β, c with $a^2 + \beta^2 \neq 0$,

$$\int e^{as}\cos(\beta s + c)\,ds = \frac{e^{as}}{a^2+\beta^2}\left[a\cos(\beta s + c) + \beta\sin(\beta s + c)\right] + C. \quad \boxed{(L)}$$

Proof by differentiation: $\frac{d}{ds}\left\{e^{as}\left[a\cos(\beta s + c) + \beta\sin(\beta s + c)\right]\right\} =$
$e^{as}\left[a^2\cos + a\beta\sin - a\beta\sin + \beta^2\cos\right] = (a^2+\beta^2)\,e^{as}\cos(\beta s + c)$.

PHASE 3

Step-by-Step Derivation

3.1 Trajectory and path line of the particle starting at the origin

Why this tool: a trajectory is the solution of $d\mathbf{x}/dt = \mathbf{v}(\mathbf{x}, t)$ with $\mathbf{x}(0) = \mathbf{x}_0$ (Notes Eq. 2.12); the path line is the curve in space obtained by eliminating t (Eq. 2.15). The system is triangular: v_y does not depend on x, so $y(t)$ is found first.

Vertical motion:

$$\frac{dy}{dt} = V \implies \int_0^y dy' = \int_0^t V\,dt' \implies y(t) = Vt.$$

Along this particle, $e^{-y/\delta} = e^{-Vt/\delta}$ and $\Theta = \Omega t - Vt/\delta = bt$. Therefore

$$\frac{dx}{dt} = U e^{-Vt/\delta}\cos(bt).$$

Apply (L) with $s = t'$, $a = -V/\delta$, $\beta = b$, $c = 0$, and Barrow's rule between the limits $t' = 0$ and $t' = t$:

$$x(t) = U\left[\frac{e^{-Vt'/\delta}}{D}\left(-\frac{V}{\delta}\cos bt' + b\sin bt'\right)\right]_{t'=0}^{t'=t}.$$

Upper limit: $\dfrac{e^{-Vt/\delta}}{D}\left(-\dfrac{V}{\delta}\cos bt + b\sin bt\right)$. Lower limit: $\dfrac{1}{D}\left(-\dfrac{V}{\delta}\cdot 1 + b\cdot 0\right) = -\dfrac{V}{\delta D}$. Subtracting,

$$\boxed{x(t) = \frac{U}{D}\left[e^{-Vt/\delta}\left(-\frac{V}{\delta}\cos bt + b\sin bt\right) + \frac{V}{\delta}\right], \qquad y(t) = Vt.}$$

Path line: substitute $t = y/V$, so that $bt = (\Omega/V - 1/\delta)\,y$:

$$x(y) = \frac{U}{D}\left[e^{-y/\delta}\left(-\frac{V}{\delta}\cos\left[\left(\frac{\Omega}{V} - \frac{1}{\delta}\right)y\right] + b\sin\left[\left(\frac{\Omega}{V} - \frac{1}{\delta}\right)y\right]\right) + \frac{V}{\delta}\right].$$

The trajectory carries the time law; the path line is a damped oscillation of the horizontal displacement versus height.

3.2 Streamline through the origin

Why this tool: a streamline is tangent to $\mathbf{v}$ at a frozen instant (Notes Eq. 2.21), so t is a parameter and not an integration variable. With t frozen:

$$\frac{dx}{v_x} = \frac{dy}{v_y} \implies \frac{dx}{dy} = \frac{U}{V} e^{-y/\delta} \cos\left(\Omega t - \frac{y}{\delta}\right).$$

Apply (L) with $s = y'$, $a = -1/\delta$, $\beta = -1/\delta$, $c = \Omega t$; here $a^2 + \beta^2 = 2/\delta^2$:

$$\int e^{-y'/\delta} \cos\Theta \, dy' = \frac{e^{-y'/\delta}}{2/\delta^2} \left[-\frac{1}{\delta}\cos\Theta - \frac{1}{\delta}\sin\Theta\right] = -\frac{\delta}{2} e^{-y'/\delta}\left(\cos\Theta + \sin\Theta\right), \qquad \Theta = \Omega t - \frac{y'}{\delta}.$$

Barrow's rule between $y' = 0$ (where $x = 0$ and $\Theta = \Omega t$) and $y' = y$:

$$\boxed{x(y,t) = \frac{U\delta}{2V}\left[\cos\Omega t + \sin\Omega t - e^{-y/\delta}\left(\cos\Theta + \sin\Theta\right)\right], \quad \Theta = \Omega t - \frac{y}{\delta}.}$$

Check by differentiation, using $\partial\Theta/\partial y = -1/\delta$: $\partial_y(\cos\Theta + \sin\Theta) = (\sin\Theta - \cos\Theta)/\delta$, so $\partial_y\left[e^{-y/\delta}(\cos\Theta + \sin\Theta)\right] = -\frac{1}{\delta}e^{-y/\delta}\left[(\cos\Theta + \sin\Theta) + (\cos\Theta - \sin\Theta)\right] = -\frac{2}{\delta}e^{-y/\delta}\cos\Theta$, hence $dx/dy = (U/V)e^{-y/\delta}\cos\Theta$ as required.

3.3 Vorticity and velocity potential

Why this tool: vorticity $\boldsymbol{\omega} = \nabla \wedge \mathbf{v}$ measures twice the local angular velocity (Notes Eq. 2.52); a velocity potential $\mathbf{v} = \nabla\varphi$ (Eq. 2.34) exists only when $\boldsymbol{\omega} \equiv 0$ in a simply connected domain.

$$\omega_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = 0 - \frac{\partial v_x}{\partial y}.$$

Chain rule, using $\partial_y e^{-y/\delta} = -e^{-y/\delta}/\delta$ and $\partial_y \cos\Theta = -\sin\Theta\,\partial_y\Theta = \sin\Theta/\delta$:

$$\frac{\partial v_x}{\partial y} = U e^{-y/\delta}\left[-\frac{1}{\delta}\cos\Theta + \frac{1}{\delta}\sin\Theta\right] \implies \boxed{\omega_z = \frac{U}{\delta} e^{-y/\delta}\left(\cos\Theta - \sin\Theta\right) \neq 0.}$$

For example, at $y = 0$ and $t = 0$: $\omega_z = U/\delta \neq 0$. The flow is rotational, so no velocity potential exists.

3.4 Rate of expansion and stream function

Why this tool: $\nabla \cdot \mathbf{v}$ is the rate of change of volume per unit volume (Notes Eqs. 2.62-2.63). If it vanishes, a planar field admits a stream function with $v_x = \partial_y \psi$, $v_y = -\partial_x \psi$ (Eq. 2.41), whose level lines are stream lines (Eq. 2.42).

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 + 0 = 0.$$

Integrate $v_y = -\partial_x \psi = V$ with respect to x: $\psi = -Vx + f(y,t)$. Then $\partial_y \psi = f'(y,t) = v_x$, and by the primitive of 3.2,

$$f(y,t) = U\int e^{-y/\delta}\cos\Theta\, dy = -\frac{U\delta}{2} e^{-y/\delta}\left(\cos\Theta + \sin\Theta\right) + \psi_0(t),$$

$$\boxed{\psi(x,y,t) = -Vx - \frac{U\delta}{2} e^{-y/\delta}\left(\cos\Theta + \sin\Theta\right) + \psi_0(t).}$$

Check: $-\partial_x\psi = V$ and $\partial_y\psi = U e^{-y/\delta}\cos\Theta = v_x$. Consistency with 3.2: along the curve of 3.2, $\psi - \psi_0 = -Vx(y,t) - \frac{U\delta}{2}e^{-y/\delta}(\cos\Theta + \sin\Theta) = -\frac{U\delta}{2}(\cos\Omega t + \sin\Omega t)$, independent of y, so that curve is a level set $\psi = \text{const}$.

3.5 Fluid line initially on the axis $x = 0$

Why this tool: a fluid line is made of particles, so it is obtained by writing the trajectories (Eq. 2.17) of all particles of the initial curve $\mathbf{x}_l(\lambda) = (0, \lambda)$ and then eliminating the label λ.

Particle λ: $y = \lambda + Vt'$ and $\Theta = \Omega t' - (\lambda + Vt')/\delta = bt' - \lambda/\delta$, so

$$\frac{dx}{dt'} = U e^{-\lambda/\delta} e^{-Vt'/\delta} \cos\left(bt' - \frac{\lambda}{\delta}\right).$$

Lemma (L) with $a = -V/\delta$, $\beta = b$, $c = -\lambda/\delta$ and Barrow's rule between $t' = 0$ and $t' = t$:

$$x(\lambda, t) = \frac{U e^{-\lambda/\delta}}{D}\left\{ e^{-Vt/\delta}\left[-\frac{V}{\delta}\cos\left(bt - \frac{\lambda}{\delta}\right) + b\sin\left(bt - \frac{\lambda}{\delta}\right)\right] - \left[-\frac{V}{\delta}\cos\frac{\lambda}{\delta} - b\sin\frac{\lambda}{\delta}\right]\right\},$$

where the lower limit used that cosine is even and sine is odd. Now eliminate $\lambda = y - Vt$ (the change of variable from label to current height). Then $e^{-\lambda/\delta} e^{-Vt/\delta} = e^{-y/\delta}$, $e^{-\lambda/\delta} = e^{-(y-Vt)/\delta}$ and $bt - \lambda/\delta = \Omega t - y/\delta = \Theta$:

$$\boxed{x(y,t) = -\frac{U e^{-y/\delta}}{D}\left[\frac{V}{\delta}\cos\Theta - b\sin\Theta\right] + \frac{U e^{-(y-Vt)/\delta}}{D}\left[\frac{V}{\delta}\cos\frac{y-Vt}{\delta} + b\sin\frac{y-Vt}{\delta}\right]}$$

with $\Theta = \Omega t - y/\delta$, $b = \Omega - V/\delta$, $D = (V/\delta)^2 + b^2$, valid for $y \geq Vt$ (the particles with $\lambda \geq 0$). At $t = 0$ it reduces to $x = 0$.

3.6 Volume flux through the plane $x = 0$

Why this tool: the volume flux through a fixed surface is $\int \mathbf{v}\cdot\mathbf{n}\, d\sigma$ (Eq. 2.36). The plane $x = 0$ has normal $\mathbf{e}_x$, so only v_x crosses it; per unit depth, $d\sigma = dy$.

$$Q(t) = \int_0^\infty v_x(0, y, t)\, dy = U\left[-\frac{\delta}{2} e^{-y/\delta}\left(\cos\Theta + \sin\Theta\right)\right]_{y=0}^{y=\infty}.$$

At the upper limit the exponential vanishes; at $y = 0$, $\Theta = \Omega t$:

$$\boxed{Q(t) = \frac{U\delta}{2}\left(\cos\Omega t + \sin\Omega t\right) = \frac{U\delta}{\sqrt{2}}\cos\left(\Omega t - \frac{\pi}{4}\right).}$$

Integral over one period, as written in the statement, with the primitive $\int(\cos\Omega t' + \sin\Omega t')\,dt' = (\sin\Omega t' - \cos\Omega t')/\Omega$ and Barrow's rule:

$$\int_t^{t+2\pi/\Omega} Q\,dt' = \frac{U\delta}{2\Omega}\left[\sin\Omega t' - \cos\Omega t'\right]_t^{t+2\pi/\Omega} = \frac{U\delta}{2\Omega}\left[(\sin\Omega t - \cos\Omega t) - (\sin\Omega t - \cos\Omega t)\right] = 0,$$

because $\Omega(t + 2\pi/\Omega) = \Omega t + 2\pi$. The time average (that integral divided by the period $2\pi/\Omega$) is also 0: there is no net volume transported through $x = 0$ over a cycle.

3.7 Flux leaving the sphere of radius R centred at $(0, 2R)$

Why this tool: for a closed surface the flux is turned into a volume integral by Gauss' formula (Eq. 2.37). The sphere (a body in the planar flow extended uniformly in z) has lowest point at $y = 2R - R = R > 0$, so it lies entirely inside the fluid domain $y > 0$ and the field is smooth in it.

$$\oint_\Sigma \mathbf{v}\cdot\mathbf{n}\,d\sigma = \int_{V_\Sigma} \nabla\cdot\mathbf{v}\,dV = \int_{V_\Sigma} 0\,dV = 0.$$

Independent check by symmetry: v_x depends on y only, so it is even under $x \to -x$ while n_x is odd, and $\oint v_x n_x\,d\sigma = 0$; the uniform $V\mathbf{e}_y$ gives $V\oint n_y\,d\sigma = 0$ for any closed surface. Result: $\boxed{0}$.

3.8 Circulation around the square

Why this tool: the circulation is $\Gamma = \oint \mathbf{v}\cdot d\mathbf{l}$ (Eq. 2.29); by Stokes (Eq. 2.32) it equals the flux of ω_z through the enclosed area provided the area lies inside the fluid. The square has vertices $(x_1, L/2)$, $(x_1 + L, L/2)$, $(x_1 + L, 3L/2)$, $(x_1, 3L/2)$ and is traversed counter-clockwise (normal $+\mathbf{e}_z$).

Line integral, side by side:

- bottom ($y = L/2$, $dx > 0$): $\int_{x_1}^{x_1+L} v_x\,dx = L\,v_x(L/2)$;
- right side ($x = x_1 + L$, $dy > 0$): $\int_{L/2}^{3L/2} V\,dy = VL$;
- top ($y = 3L/2$, $dx < 0$): $-L\,v_x(3L/2)$;
- left side ($dy < 0$): $-VL$.

The vertical contributions cancel, hence

$$\boxed{\Gamma = UL\left[e^{-L/2\delta}\cos\left(\Omega t - \frac{L}{2\delta}\right) - e^{-3L/2\delta}\cos\left(\Omega t - \frac{3L}{2\delta}\right)\right].}$$

Stokes theorem:

$$\Gamma = \int_{x_1}^{x_1+L}\int_{L/2}^{3L/2}\omega_z\,dy\,dx = L\int_{L/2}^{3L/2}\left(-\frac{\partial v_x}{\partial y}\right)dy = -L\left[v_x\right]_{y=L/2}^{y=3L/2} = L\left[v_x\left(\tfrac{L}{2}\right) - v_x\left(\tfrac{3L}{2}\right)\right],$$

which is the same expression.

PHASE 4

Interpretation, Limits and Dimensional Check

- Wall limit: at $y = 0$, $v_x = U\cos\Omega t$ (the wall velocity) and the oscillation decays as $e^{-y/\delta}$, with a phase lag y/δ. For $y \gg \delta$ the fluid has no motion in x and only the uniform blowing V remains.
- Limit $V \to 0$ in 3.1: $D \to \Omega^2$, $b \to \Omega$ and $x(t) \to (U/\Omega)\sin\Omega t$, the displacement of the wall itself, as expected for a particle that stays on the wall.
- Limit $V \to 0$ in 3.2: $x \propto 1/V \to \infty$, i.e. the streamlines become parallel to the wall, consistent with $v_y \to 0$.
- The flow is incompressible and rotational; it has a stream function but no potential, as found in 3.3 and 3.4. The zero net flux in 3.7 and the zero cycle integral in 3.6 follow from this solenoidal, periodic character.
- Dimensions (SI): $[x(t)] = [U/D]\,[V/\delta] = (\mathrm{m/s})(\mathrm{s^2})(\mathrm{s^{-1}}) = \mathrm{m}$. $[\omega_z] = [U/\delta] = \mathrm{s^{-1}}$. $[\psi] = [Vx] = [U\delta] = \mathrm{m^2/s}$. $[Q] = [U\delta] = \mathrm{m^2/s}$ per unit depth. $[\Gamma] = [UL] = \mathrm{m^2/s}$.
- Numerical cross-check (quadrature of the defining integral against the closed form), with $U = 1.3$ m/s, $V = 0.7$ m/s, $\delta = 0.9$ m, $\Omega = 2.1$ s^{-1}, $t = 1.7$ s: path $x(t) = 0.653175$ m (both); streamline at $y = 0.8$ m, $x = -0.952255$ m (both); fluid line (valid only for $y \geq Vt = 1.19$ m) at $y = 1.5$ m, $x = 0.618267$ m, and at $y = 2.5$ m, $x = 0.194357$ m (both); $Q = -0.775155$ m^2/s (both); for $L = 0.6$ m, $\Gamma = -0.315037$ m^2/s from the line integral, from the closed form and from the Stokes integral.

Reading of the official handwritten solution

In K1.pdf the first bracket of part 5 is written $\frac{V}{\delta}\cos[\cdot] - (\Omega - \frac{V}{\delta})\sin[\cdot]$ and, as legible in the scan, the argument of that sine shows y where t is meant. The expression above is the result of the explicit Barrow derivation. The statement of part 6 defines Q_a as the bare integral over one period (no factor $\Omega/2\pi$); the handwritten answer is 0, which agrees with the integral computed above, and the mean value is also 0.

PROBLEM K2

Couette flow | Vorticity | Strain rate | Fluid element

Plane Couette Flow, Rotation and Strain

Original statement · K2.pdf

OFFICIAL PROBLEM STATEMENT

Consider the steady flow of a liquid confined between two parallel planar walls separated a distance H. If the top wall moves with constant velocity U, the resulting fluid motion exhibits a single velocity component $v_x = Uy/H$, where y denotes the distance to the bottom wall, assumed to be at rest.

1. Determine the streamlines and the trajectories.
2. Obtain the vorticity field.
3. Compute the circulation Γ around a rectangular line of side L oriented parallel to the walls and height h.
4. Calculate the gradient of velocity, the rate-of-strain tensor and the expansion rate.
5. Obtain the principal directions of strain as well as the associated principal strain rates.
6. Investigate the evolution of a square fluid element of side dl oriented parallel to the walls.

Source: K2.pdf (page 1). Notes.pdf, Chapter 2: trajectories Eq. (2.12), stream lines Eq. (2.21), circulation and Stokes theorem Eqs. (2.29)-(2.33), velocity gradient and its decomposition Eqs. (2.46)-(2.52), principal directions Eqs. (2.54)-(2.55), square element Eqs. (2.56)-(2.61), expansion rate Eqs. (2.62)-(2.63).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar, steady, unidirectional flow: $\mathbf{v} = (Uy/H)\,\mathbf{e}_x$, with $v_y = v_z = 0$. The velocity depends on one coordinate only (y), so there is a single spatial degree of freedom; the time derivative of the field vanishes.
- The domain is $0 \le y \le H$ (liquid between the walls); the bottom wall ($y = 0$) is at rest and the top wall ($y = H$) moves with velocity U, so the no-slip condition is satisfied at both walls.
- The liquid is treated as a continuum of constant density, as stated ("a liquid").

Symbol	Meaning	SI unit
U	velocity of the top wall	m/s
H	gap between the walls	m
L, h	length (along x) and height (along y) of the rectangular circuit	m
dl	side of the fluid element	m
U/H	shear rate	s^{-1}

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial laboratory frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, $\mathbf{e}_z = \mathbf{e}_x \wedge \mathbf{e}_y$ (out of the plane of the motion). The bottom wall defines the frame at rest.
- Convention for the velocity gradient (Notes, after Eq. 2.26): $(\nabla \mathbf{v})_{ij} = \partial v_j / \partial x_i$, so that the change of the velocity across a small material segment $d\mathbf{x}$ is the row vector times the matrix, $d\mathbf{v} = d\mathbf{x} \cdot \nabla \mathbf{v}$ (Eq. 2.46).
- The principal axes of strain are obtained from a change of basis: the eigenvectors of the symmetric tensor $\bar{\bar{T}}_d$ give a rotated basis $(\mathbf{n}_1, \mathbf{n}_2)$ at $\pm 45^\circ$ to $(\mathbf{e}_x, \mathbf{e}_y)$ (part 5). The change-of-basis matrix has the vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ as columns.

PHASE 3

Step-by-Step Derivation

3.1 Streamlines and trajectories

Why this tool: streamlines are tangent to $\mathbf{v}$ at one instant (Eq. 2.21); trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12). For a steady field both are determined by the same vector field and, as shown below, they coincide geometrically.

Streamlines: $dx/v_x = dy/v_y$ with $v_y = 0$. Written as $v_y\,dx - v_x\,dy = 0$ this reads $-v_x\,dy = 0$, and $dy = 0$ wherever $v_x \neq 0$. Hence $y = y_0 =$ const: straight lines parallel to the walls.

Trajectories with $\mathbf{x}(0) = (x_0, y_0)$:

$$\frac{dy}{dt} = 0 \implies y(t) = y_0, \qquad \frac{dx}{dt} = \frac{U y_0}{H} \implies \int_{x_0}^{x} dx' = \int_0^t \frac{U y_0}{H}\,dt' \implies x(t) = x_0 + \frac{U y_0}{H}\,t.$$

Eliminating t gives the path line $y = y_0$. The particles on the bottom wall are at rest ($y_0 = 0$) and the particles at $y_0 = H$ move at U. Streamlines, path lines and (because the field is steady) streak lines are the same family of lines $y =$ const.

3.2 Vorticity

Why this tool: $\boldsymbol{\omega} = \nabla \wedge \mathbf{v}$ is twice the angular velocity of the material element (Eq. 2.52).

$$\omega_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = 0 - \frac{\partial}{\partial y}\left(\frac{U y}{H}\right) = -\frac{U}{H}, \qquad \boxed{\boldsymbol{\omega} = -\frac{U}{H}\,\mathbf{e}_z.}$$

The vorticity is uniform and non-zero: the flow is rotational, with clockwise rotation (the top moves in $+x$, the bottom is at rest).

3.3 Circulation around a rectangle

Why this tool: by definition $\Gamma = \oint \mathbf{v}\cdot d\mathbf{l}$ (Eq. 2.29); since the rectangle lies inside the fluid, Stokes theorem (Eq. 2.32) gives the same value from the vorticity flux. The circuit occupies $x \le x' \le x + L$ and $y \le y' \le y + h$ with $0 \le y$ and $y + h \le H$, and is traversed counter-clockwise (normal $+\mathbf{e}_z$).

Line integral, bottom + right - top - left:

$$\Gamma = \int_x^{x+L} v_x(y)\,dx' + \int_y^{y+h} v_y\,dy' - \int_x^{x+L} v_x(y+h)\,dx' - \int_y^{y+h} v_y\,dy'.$$

Both vertical integrals vanish because $v_y \equiv 0$, and the horizontal ones are integrals of constants over the length L:

$$\Gamma = \frac{Uy}{H}L - \frac{U(y+h)}{H}L = \boxed{-\frac{UhL}{H}.}$$

Stokes theorem: $\Gamma = \iint_S \boldsymbol{\omega}\cdot\mathbf{e}_z\,d\sigma = \left(-\frac{U}{H}\right)(hL)$, which is the same.

3.4 Velocity gradient, rate-of-strain tensor and expansion rate

Why this tool: splitting $\nabla\mathbf{v}$ into symmetric and anti-symmetric parts (Eq. 2.47) separates the pure strain of the element, $\bar{\bar{T}}_d$, from its rigid rotation, $\bar{\bar{T}}_r$.

$$\nabla\mathbf{v} = \begin{pmatrix} \partial_x v_x & \partial_x v_y \\ \partial_y v_x & \partial_y v_y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ U/H & 0 \end{pmatrix},$$

$$\bar{\bar{T}}_d = \tfrac{1}{2}\left[\nabla\mathbf{v} + (\nabla\mathbf{v})^T\right] = \begin{pmatrix} 0 & \dfrac{U}{2H} \\ \dfrac{U}{2H} & 0 \end{pmatrix}, \qquad \bar{\bar{T}}_r = \tfrac{1}{2}\left[\nabla\mathbf{v} - (\nabla\mathbf{v})^T\right] = \begin{pmatrix} 0 & -\dfrac{U}{2H} \\ \dfrac{U}{2H} & 0 \end{pmatrix}.$$

Expansion rate (the trace, Eqs. 2.62-2.63): $\nabla\cdot\mathbf{v} = \operatorname{tr}\bar{\bar{T}}_d = 0 + 0 = 0$. The liquid element keeps its volume.

Consistency with the vorticity: in this convention $(\bar{\bar{T}}_r)_{12} = \frac{1}{2}\left(\partial_x v_y - \partial_y v_x\right) = \omega_z/2 = -U/2H$ is the angular velocity of the element, and $d\mathbf{v}_r = d\mathbf{x}\cdot\bar{\bar{T}}_r = \frac{1}{2}\boldsymbol{\omega}\wedge d\mathbf{x}$ (Eq. 2.52). Check with $d\mathbf{x} = (0, dl)$: $d\mathbf{x}\cdot\bar{\bar{T}}_r = (dl\,U/2H,\,0)$ and $\frac{1}{2}(-U/H)\,\mathbf{e}_z\wedge dl\,\mathbf{e}_y = +\frac{U}{2H}dl\,\mathbf{e}_x$; both agree.

3.5 Principal directions and strain rates

Why this tool: along the principal directions the strain reduces to a pure extension without shear (Eq. 2.54), so the eigenproblem of the symmetric tensor $\bar{\bar{T}}_d$ is solved through its characteristic equation (Eq. 2.55).

$$\det\left(\bar{\bar{T}}_d - \lambda\bar{\bar{I}}\right) = \begin{vmatrix} -\lambda & U/2H \\ U/2H & -\lambda \end{vmatrix} = \lambda^2 - \left(\frac{U}{2H}\right)^2 = 0 \implies \lambda_{1,2} = \pm\frac{U}{2H}.$$

For $\lambda_1 = +U/2H$: $\begin{pmatrix} -U/2H & U/2H \\ U/2H & -U/2H \end{pmatrix}\begin{pmatrix} n_x \\ n_y \end{pmatrix} = 0 \Rightarrow n_x = n_y$, and normalising, $\mathbf{n}_1 = \frac{1}{\sqrt{2}}(1,1)$ (angle $+45^\circ$).

For $\lambda_2 = -U/2H$: $\begin{pmatrix} U/2H & U/2H \\ U/2H & U/2H \end{pmatrix}\begin{pmatrix} n_x \\ n_y \end{pmatrix} = 0 \Rightarrow n_x = -n_y$, so $\mathbf{n}_2 = \frac{1}{\sqrt{2}}(1,-1)$ (angle -45°).

Change of basis: with $\mathbf{P} = [\mathbf{n}_1\ \mathbf{n}_2] = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ we have $\mathbf{P}\mathbf{P}^T = \mathbf{I}$ and $\det\mathbf{P} = -1$. The pair $(\mathbf{n}_1, \mathbf{n}_2)$ is orthonormal but left-handed; using $-\mathbf{n}_2$ instead gives a proper rotation with $\det = +1$. In either case $\mathbf{P}^T\bar{\bar{T}}_d\mathbf{P} = \mathrm{diag}(U/2H, -U/2H)$.

The material is stretched along $+45^\circ$ and compressed along -45°, at the rate $U/2H$ per unit length in each direction; the sum of the principal rates is zero, as required by $\nabla\cdot\mathbf{v} = 0$.

3.6 Evolution of a square fluid element

Why this tool: a material segment $d\mathbf{l}$ evolves as $d\mathbf{l}(t+dt) = d\mathbf{l} + (d\mathbf{l}\cdot\nabla\mathbf{v})\,dt$ (Eqs. 2.46 and 2.56-2.57), valid to first order in dt.

Horizontal side, $d\mathbf{l}_1 = dl\,(1,0)$: $d\mathbf{l}_1\cdot\nabla\mathbf{v} = dl\,(1,0)\begin{pmatrix} 0 & 0 \\ U/H & 0 \end{pmatrix} = dl\,(0,0)$, so $d\mathbf{l}_1 \to dl\,(1,0)$: it neither stretches nor turns.

Vertical side, $d\mathbf{l}_2 = dl\,(0,1)$: $d\mathbf{l}_2\cdot\nabla\mathbf{v} = dl\,(0,1)\begin{pmatrix} 0 & 0 \\ U/H & 0 \end{pmatrix} = dl\,(U/H,\,0)$, so

$$d\mathbf{l}_2 \to dl\begin{pmatrix} \dfrac{U}{H}\,dt,\ 1 \end{pmatrix}.$$

The square becomes a parallelogram: the vertical side tilts towards $+x$ by the small angle $(U/H)\,dt$, and the angle between the two sides decreases from $\pi/2$ to $\pi/2 - (U/H)\,dt$. The area, base dl times height dl, is unchanged to first order (zero expansion rate).

Decomposition of the same motion (Eq. 2.51 and 2.60-2.61), side by side:

- Horizontal side: $d\mathbf{l}_1 \cdot \bar{\bar{T}}_r = dl\,(0, -U/2H)$ (rotation, clockwise) and $d\mathbf{l}_1 \cdot \bar{\bar{T}}_d = dl\,(0, +U/2H)$ (strain, counter-clockwise); the two cancel, as found directly.
- Vertical side: $d\mathbf{l}_2 \cdot \bar{\bar{T}}_r = dl\,(U/2H, 0)$ and $d\mathbf{l}_2 \cdot \bar{\bar{T}}_d = dl\,(U/2H, 0)$; both move the tip towards $+x$ and add up to $dl\,(U/H, 0)$.

So the distortion is the combined result of a rigid rotation with angular velocity $\Omega_z = -U/2H$ (angle $-(U/2H)\,dt$) and a shear strain with strain rate $\dot{\gamma} = \partial_x v_y + \partial_y v_x = U/H$, which is the rate at which the angle between the sides decreases (Notes, after Eq. 2.61).

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: layers slide over each other with uniform shear rate U/H; rotation and strain contribute equally, $\omega_z = -\dot{\gamma}$, so that $|\Omega_z| = \dot{\gamma}/2$. The principal strain axes are at $\pm 45^\circ$, as in any simple shear.
- Limits: $U \to 0$ gives rest ($\boldsymbol{\omega}$, $\bar{\bar{T}}_d$ and Γ vanish); $H \to \infty$ at fixed U gives vanishing shear rate. For $h \to 0, \Gamma \to 0$ linearly and $\Gamma/(hL) = -U/H = \omega_z$, which is Eq. (2.33): the circulation per unit area equals the normal component of the vorticity.
- Incompressibility: $\nabla \cdot \mathbf{v} = 0, \operatorname{tr} \bar{\bar{T}}_d = 0$ and $\lambda_1 + \lambda_2 = 0$ are mutually consistent.
- Dimensions (SI): $[U/H] = \mathrm{s}^{-1}$ for ω_z, $\nabla\mathbf{v}$, $\bar{\bar{T}}_d$ and the principal strain rates; $[\Gamma] = [UhL/H] = (\mathrm{m/s})(\mathrm{m})(\mathrm{m})/\mathrm{m} = \mathrm{m}^2/\mathrm{s}$; $[x(t)] = [Uy_0t/H] = \mathrm{m}$.
- Symbolic cross-check: for the gradient above, the eigen-decomposition of $\bar{\bar{T}}_d$ returns eigenvalue $+U/2H$ with eigenvector $(1, 1)$ and $-U/2H$ with eigenvector $(-1, 1)$, which is $\mathbf{n}_2$ up to sign.

READING OF THE OFFICIAL HANDWRITTEN SOLUTION

K2.pdf states the final decomposition as a rotation with angular velocity $-U/2H$ plus a shear strain rate U/H, which agrees with the result above, and writes $\nabla\mathbf{v}$ with the same convention $(\nabla\mathbf{v})_{ij} = \partial_i v_j$ used here. No discrepancy was found in this problem.

PROBLEM K3

Spherical coordinates | Fluid surface | Stream tube | Moving observer

Three-Dimensional Point Source at the Origin

Original statement · K3.pdf

OFFICIAL PROBLEM STATEMENT

A three-dimensional fluid source placed at the origin induces a radial motion given by $v_r = Q(t)/(4\pi r^2)$, with $v_\theta = v_\phi = 0$. Obtain:

1. Streamlines, trajectories and paths.
2. Equation for the fluid surface initially located at $r = R$.
3. Equation for the fluid line that corresponds at the initial instant with a circle of radius R intersecting the origin.
4. Streamtube that intersects the circle defined by $r = R_1$ and $\theta = \theta_1$.
5. The velocity components in cartesian coordinates, as well as the corresponding streamlines and trajectories.
6. Equations for the streamlines and trajectories for an observer moving with velocity U.

Source: K3.pdf (page 1). Notes.pdf, Chapter 2: spherical coordinates and scale factors (Fig. 2.1, Eqs. 2.3-2.7), trajectories Eq. (2.12), fluid lines and surfaces Eqs. (2.16)-(2.20), stream lines and stream tubes Eq. (2.21), observer dependence of steadiness (text before Eq. 2.11), convective flux Eqs. (2.35)-(2.38).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Spherically symmetric, purely radial flow: $\mathbf{v} = \dfrac{Q(t)}{4\pi r^2}\mathbf{e}_r$ for $r > 0$. The field depends on r and t only. The point $r = 0$ is the source and is excluded from the domain.
- $Q(t)$ is the volume flux emitted by the source (volume per unit time) and is a given function of time, so the flow is unsteady in general.
- The accumulated emitted volume is $q(t) \equiv \int_0^t Q(t')\,dt'$ (volume). We assume $r^3 > 0$ for all particles considered (automatically true for $Q \geq 0$).
- Each particle has three degrees of freedom (r, θ, ϕ); the field acts on r only, so θ and ϕ are conserved by each particle.

Symbol	Meaning	SI unit
$Q(t)$	source strength (volume flux)	$\mathrm{m^3/s}$
$q(t)$	emitted volume	$\mathrm{m^3}$
r_0, R, R_1	initial radius, radius of the initial surface/circle, radius of the circle of part 4	m
θ, ϕ	polar angle (from $+z$) and azimuth	rad
U	speed of the observer along x	m/s

PHASE 2

Coordinates, Frames and Changes of Variable

- Spherical coordinates (r, θ, ϕ) with polar axis z and scale factors $(1, r, r\sin\theta)$ (Notes, Section on coordinate systems): $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$, $r = (x^2+y^2+z^2)^{1/2}$.
- Basis change used in part 5: $\mathbf{e}_r = (x\,\mathbf{e}_x + y\,\mathbf{e}_y + z\,\mathbf{e}_z)/r$.
- Frames: the laboratory frame S (source at rest at the origin) for parts 1-5; for part 6 an inertial frame S' moving with constant velocity $U\mathbf{e}_x$ relative to S, with axes parallel and coincident origins at $t = 0$, so $x' = x - Ut$, $y' = y$, $z' = z$ and $\mathbf{v}' = \mathbf{v} - U\mathbf{e}_x$ (Galilean change of observer; no inertial terms because S' is not accelerated, cf. Eq. 2.28).
- Lagrangian labels: initial position (r_0, θ_0, ϕ_0).

PHASE 3

Step-by-Step Derivation

3.1 Streamlines, trajectories and paths

Why this tool: streamlines come from the tangency condition (Eq. 2.21, here with scale factors $1, r, r\sin\theta$); trajectories from the initial-value problem $d\mathbf{x}/dt = \mathbf{v}$ (Eqs. 2.12-2.13).

Streamlines at a frozen instant:

$$\frac{dr}{v_r} = \frac{r\,d\theta}{v_\theta} = \frac{r\sin\theta\,d\phi}{v_\phi}.$$

With $v_\theta = v_\phi = 0$ the second and third ratios are only compatible with $d\theta = 0$ and $d\phi = 0$, i.e. $\theta = \theta_0$ and $\phi = \phi_0$: the streamlines are straight rays through the origin, at every instant.

Trajectories, from (r_0, θ_0, ϕ_0) at $t = 0$:

$$r\frac{d\theta}{dt} = 0, \quad r\sin\theta\frac{d\phi}{dt} = 0 \implies \theta(t) = \theta_0,\ \phi(t) = \phi_0, \qquad \frac{dr}{dt} = \frac{Q(t)}{4\pi r^2}.$$

Separate variables and integrate with Barrow's rule between $(r_0, 0)$ and (r, t):

$$\int_{r_0}^{r} 4\pi r'^2\,dr' = \int_0^t Q(t')\,dt' \implies \frac{4\pi}{3}\left[r'^3\right]_{r_0}^{r} = q(t) \implies \boxed{\frac{4\pi}{3}\left(r^3 - r_0^3\right) = q(t) = \int_0^t Q\,dt',}$$

$$r(t) = \left[r_0^3 + \frac{3}{4\pi}\,q(t)\right]^{1/3}.$$

Because θ and ϕ never change, the path lines are the same rays as the streamlines, but the particle moves along them with the time law above. The physical content of the boxed equation is volume conservation: the sphere of radius r contains the sphere of radius r_0 plus the emitted volume.

3.2 Fluid surface initially at $r = R$

Why this tool: a fluid surface is the set of trajectories of the particles of the initial surface (Eqs. 2.18-2.20). The initial surface is $\mathbf{x}_s(\alpha,\beta) = (r_0 = R,\ \theta_0 = \alpha,\ \phi_0 = \beta)$ with $\alpha \in [0,\pi]$, $\beta \in [0, 2\pi)$.

Apply 3.1 with $r_0 = R$ and $\theta = \alpha$, $\phi = \beta$:

$$\frac{4\pi}{3}\left(r^3 - R^3\right) = q(t), \quad \theta = \alpha, \quad \phi = \beta.$$

The parameters α, β are the spherical angles themselves, so eliminating them leaves a relation in r and t only:

$$\boxed{r(t) = \left[R^3 + \frac{3}{4\pi}\int_0^t Q\,dt'\right]^{1/3}.}$$

The fluid surface remains a sphere centred at the source; the volume it encloses grows exactly by the emitted volume $q(t)$.

3.3 Fluid line initially a circle of radius R through the origin

Why this tool: the fluid line is again the image of the initial curve under the trajectories (Eq. 2.17); the label of the curve is eliminated at the end.

Initial curve. Take the circle in the plane $\phi = 0$ (the x-z plane), tangent to the x-axis at the origin, with centre $(x, z) = (0, R)$. In polar form this circle is $r_0 = 2R\cos\lambda$, where $\lambda \in (-\pi/2, \pi/2)$ is the signed polar angle from the z-axis (negative λ labels the half-plane $\phi = \pi$). Check: $x_0 = r_0 \sin\lambda = R\sin 2\lambda$ and $z_0 = r_0\cos\lambda = R(1+\cos 2\lambda)$, so $x_0^2 + (z_0 - R)^2 = R^2\left(\sin^2 2\lambda + \cos^2 2\lambda\right) = R^2$. The origin corresponds to $\lambda \to \pm\pi/2$, $r_0 \to 0$.

Evolution: from 3.1, $\theta = \lambda$, ϕ unchanged and

$$\frac{4\pi}{3}\left(r^3 - 8R^3\cos^3\lambda\right) = q(t).$$

Eliminate the label with $\lambda = \theta$ (the polar angle of the particle):

$$\boxed{\frac{4\pi}{3}\left(r^3 - 8R^3\cos^3\theta\right) = \int_0^t Q\,dt' \iff r(\theta,t) = \left[8R^3\cos^3\theta + \frac{3}{4\pi}\int_0^t Q\,dt'\right]^{1/3}, \qquad |\theta| < \tfrac{\pi}{2}.}$$

At $t = 0$ it is $r = 2R\cos\theta$, the initial circle; for $t > 0$ the line is dilated, and the particle that started at the source ($\lambda = \pm\pi/2$) lies on the sphere $r = [3q/4\pi]^{1/3}$ in the plane of the circle.

> **Erratum in the official solution**
>
> The handwritten K3.pdf (part 3) takes $r_0 = R\cos\lambda$ and writes $\frac{4\pi}{3}(r^3 - R^3\cos^3\theta) = \int_0^t Q\,dt'$. The polar equation $r = R\cos\theta$ is a circle of diameter R (radius $R/2$), not of radius R; a circle of radius R through the origin is $r = 2R\cos\theta$, which gives the factor $8R^3$ above. The two results coincide only after the substitution $R \to 2R$.

3.4 Streamtube through the circle $r = R_1$, $\theta = \theta_1$

Why this tool: a stream tube is the surface formed by the streamlines that cross a closed curve (Notes, after Eq. 2.21). The curve $r = R_1, \theta = \theta_1$ ($0 \le \phi < 2\pi$) is a horizontal circle of radius $R_1\sin\theta_1$ about the polar axis.

Each streamline through a point of that circle is the ray $\theta = \theta_1, \phi = \phi_0$ (3.1). The union over $\phi_0 \in [0, 2\pi)$ and all $r > 0$ gives

$$\boxed{\theta = \theta_1,}$$

a right circular cone of half-angle θ_1 with its vertex at the source. The value of R_1 does not matter: any circle $\theta = \theta_1$ generates the same cone.

Flux inside the tube. The solid angle of the cone is $2\pi(1 - \cos\theta_1)$, and $\mathbf{v} \cdot \mathbf{e}_r \, r^2 = Q/4\pi$ is independent of r, so the volume flux through any spherical cap of the cone is

$$\int \mathbf{v} \cdot \mathbf{n} \, d\sigma = \frac{Q}{4\pi r^2} \cdot r^2 \, 2\pi(1 - \cos\theta_1) = \frac{Q}{2}\,(1 - \cos\theta_1)\,,$$

which is constant along the tube, as it must be since no fluid crosses a stream tube. For $\theta_1 = \pi$ it gives Q.

3.5 Cartesian components, streamlines and trajectories

Why this tool: the same field is written in the Cartesian basis, where the trajectory equations are three coupled equations that can be decoupled using the radial symmetry.

Using $\mathbf{e}_r = (x, y, z)/r$ and $v_r = Q/(4\pi r^2)$,

$$\boxed{v_x = \frac{Q(t)}{4\pi}\frac{x}{(x^2+y^2+z^2)^{3/2}}, \quad v_y = \frac{Q(t)}{4\pi}\frac{y}{(x^2+y^2+z^2)^{3/2}}, \quad v_z = \frac{Q(t)}{4\pi}\frac{z}{(x^2+y^2+z^2)^{3/2}}.}$$

Streamlines (frozen t): the common factor $Q/4\pi r^3$ cancels in

$$\frac{dx}{v_x} = \frac{dy}{v_y} = \frac{dz}{v_z} \implies \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z} \implies \ln\frac{|x|}{|x_0|} = \ln\frac{|y|}{|y_0|} = \ln\frac{|z|}{|z_0|},$$

$$\boxed{\frac{x}{x_0} = \frac{y}{y_0} = \frac{z}{z_0},}$$

the rays through the origin and the point (x_0, y_0, z_0).

Trajectories: from the streamline relations the position vector stays proportional to its initial value, so $\rho \equiv x/x_0 = y/y_0 = z/z_0 > 0$ is a common factor (assume $x_0 \neq 0$; otherwise use a non-zero component) and $r = \rho\, r_0$. Then

$$\frac{dx}{dt} = \frac{Q}{4\pi}\frac{x}{r^3} = \frac{Q}{4\pi}\frac{x_0^3}{r_0^3}\frac{1}{x^2} \implies \int_{x_0}^{x} x'^2\, dx' = \frac{x_0^3}{4\pi r_0^3}\int_0^t Q\, dt' \implies \frac{x^3 - x_0^3}{3} = \frac{x_0^3}{4\pi r_0^3}\, q(t),$$

where $x = \rho x_0$ and $r = \rho r_0$ were used in x/r^3. Hence $x^3 = x_0^3\left[1 + 3q/(4\pi r_0^3)\right]$, that is $\rho^3 = r^3/r_0^3 = 1 + 3q/(4\pi r_0^3)$, which is the law of 3.1, and $\mathbf{x}(t) = (r(t)/r_0)\,\mathbf{x}_0$.

3.6 Observer moving with velocity U

Why this tool: steadiness depends on the observer (Notes, text before Eq. 2.11), so the flow lines of the source seen from S' must be recomputed from the transformed field, with the lab position written in terms of the observer coordinates. The source is fixed in S (the statement gives the flow induced by a source placed at the origin of the lab), so in S' it sits at $x' = -Ut$.

With $x = x' + Ut, y = y', z = z'$, define $\xi \equiv x' + Ut$ and $\rho_s^2 \equiv \xi^2 + y'^2 + z'^2$. Then

$$v'_x = \frac{Q(t)}{4\pi}\frac{\xi}{\rho_s^3} - U, \qquad v'_y = \frac{Q(t)}{4\pi}\frac{y'}{\rho_s^3}, \qquad v'_z = \frac{Q(t)}{4\pi}\frac{z'}{\rho_s^3}.$$

The relative field depends explicitly on t through Ut and $Q(t)$ even when Q is constant, so it is unsteady.

Trajectories relative to the observer. A fluid particle has lab trajectory $\mathbf{x}(t) = (r(t)/r_0)\,\mathbf{x}_0$ (3.5). Its position in S' is $\mathbf{x}'(t) = \mathbf{x}(t) - Ut\,\mathbf{e}_x$:

$$\boxed{\mathbf{x}'(t) = r(t)\,\mathbf{e}_r(\theta_0, \phi_0) - Ut\,\mathbf{e}_x, \qquad r(t) = \left[r_0^3 + \tfrac{3}{4\pi}q(t)\right]^{1/3}.}$$

These curves are generally curved (a radial motion with decreasing speed plus a uniform drift), and satisfy $d\mathbf{x}'/dt = \mathbf{v}'(\mathbf{x}', t)$ by construction.

Streamlines at a frozen instant t. The relative field is axisymmetric about the x'-axis through the source. Write $\varpi^2 = y'^2 + z'^2$. Away from the source $\nabla \cdot \mathbf{v}' = 0$, and for such axisymmetric fields the Stokes stream function Ψ with $v'_\xi = \varpi^{-1}\partial_\varpi \Psi$, $v'_\varpi = -\varpi^{-1}\partial_\xi \Psi$ exists; then $d\Psi = -\varpi\left(v'_\varpi\, d\xi - v'_\xi\, d\varpi\right) = 0$ along streamlines. For the source plus the uniform stream $-U\mathbf{e}_x$,

$$\Psi = -\frac{Q(t)}{4\pi}\frac{\xi}{\rho_s} - \frac{U}{2}\varpi^2,$$

which reproduces $v'_\xi = \frac{Q}{4\pi}\xi/\rho_s^3 - U$ and $v'_\varpi = \frac{Q}{4\pi}\varpi/\rho_s^3$ (checked symbolically). The instantaneous streamlines are therefore the curves, in each meridian plane $\phi' = \text{const}$,

$$\boxed{\frac{Q(t)}{4\pi}\frac{\xi}{\rho_s} + \frac{U}{2}\left(y'^2 + z'^2\right) = \text{const}, \qquad \xi = x' + Ut,}$$

equivalently the ODE system $dx'/v'_x = dy'/v'_y = dz'/v'_z$ at frozen t. On the axis ($\varpi = 0$, $\xi > 0$) the velocity vanishes at $\xi_s = \sqrt{Q/(4\pi U)}$ (a stagnation point, Eq. 2.11), and the streamline through it, $\frac{Q}{4\pi}\left(1 - \xi/\rho_s\right) = \frac{U}{2}\varpi^2$, is the instantaneous half-body of revolution, whose far-wake radius $\varpi_\infty = \sqrt{Q/(\pi U)}$ satisfies $Q = U\pi\varpi_\infty^2$.

Unfinished official solution

K3.pdf stops at part 6 with the note "to be continued" and writes $v'_x = \frac{Q}{4\pi}\frac{x}{(x^2+y^2+z^2)^{3/2}} - U$ with unshifted coordinates. That expression is correct only if the source moves together with the observer (source at $x' = 0$), which is a different, steady problem (a point source in a uniform stream, a half-body of revolution for constant Q). For a source fixed in the laboratory, the source sits at $x' = -Ut$ and the field is the shifted, unsteady one derived above.

PHASE 4

Interpretation, Limits and Dimensional Check

- Mass conservation: $\nabla \cdot \mathbf{v} = r^{-2}\,\partial_r\left(r^2 v_r\right) = r^{-2}\,\partial_r[Q/4\pi] = 0$ for $r > 0$, and the flux through any sphere centred at the source is $4\pi r^2 v_r = Q$, independent of r (the whole emitted flux). The fluid surface of 3.2 and the equation $\frac{4\pi}{3}(r^3 - R^3) = q$ express the

same fact.

- Late-time behaviour: for constant Q, $r \simeq (3Qt/4\pi)^{1/3}$, so the radial speed $dr/dt = Q/(4\pi r^2) \propto t^{-2/3}$ decays; for $t \to 0$ and $r_0 > 0$, $r \to r_0$.
- Observer limit: for $U \to 0$ the quantities of 3.6 reduce to those of 3.1-3.5 (streamlines $\frac{\xi}{\rho_s} = \text{const}$, i.e. rays).
- Dimensions (SI): $[v_r] = [Q/r^2] = (\mathrm{m^3/s})/\mathrm{m^2} = \mathrm{m/s}$; $[q] = \mathrm{m^3}$ and $[\tfrac{4\pi}{3} r^3] = \mathrm{m^3}$; $[\xi_s] = [\sqrt{Q/U}] = \sqrt{\mathrm{m^3/s \cdot s/m}} = \mathrm{m}$; $[\Psi] = [Q/4\pi] = [U\varpi^2] = \mathrm{m^3/s}$.
- Numerical cross-check (RK4 integration of the particle ODE against the closed form), with $Q(t) = Q_0(1 + 0.5\sin t)$, $Q_0 = 2.0$ m^3/s, $U = 0.7$ m/s, initial point $(0.8, 0.5, 0.3)$ m, $t = 3$ s: lab radius 1.4223654 m (both) with the direction cosines unchanged; relative position $(-0.950555,\ 0.718403,\ 0.431042)$ m from the ODE in S' and from the formula of 3.6 (agreement to 10^{-11} m). The circle of 3.3 satisfies $x_0^2 + (z_0 - R)^2 = R^2$ to machine precision for $R = 1.3$ m, $\lambda = 0.7$. The Stokes stream function reproduces the components of $\mathbf{v}'$ exactly (symbolic).

PROBLEM K4

Potential flow | Stagnation point | Stream function | Linearisation

Source in a Uniform Stream (Rankine Half-Body)

Original statement · K4.pdf

OFFICIAL PROBLEM STATEMENT

Consider the fluid flow that derives from the potential

$$\varphi = \frac{Q}{2\pi} \ln r + U_\infty r \cos\theta,$$

corresponding to the superposition of a uniform stream of velocity U_∞ and a source of volumetric flux Q.

1. Determine the velocity components in cartesian and cylindrical coordinates.
2. Obtain the stream lines.
3. Compute the trajectories and paths.
4. Calculate the stagnation points, sketching the stream lines that reach them.
5. Consider the flow near the stagnation points, determining in particular the trajectories and the stream lines.

Source: K4.pdf (page 1). Notes.pdf, Chapter 2: potential Eq. (2.34), gradient in orthogonal coordinates Eq. (2.3), stagnation points Eq. (2.11), trajectories Eqs. (2.12)-(2.15), stream lines Eq. (2.21), stream function Eqs. (2.40)-(2.45), velocity gradient Eq. (2.46).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar flow in the plane (r, θ) (or (x, y)), steady, derived from a potential: $\mathbf{v} = \nabla\varphi$. The flow is irrotational and, since $\nabla^2\varphi = 0$ for $r > 0$, incompressible.
- The point $r = 0$ is the source and is excluded from the domain. Because the problem is two-dimensional, Q is the volume flux per unit length normal to the plane.
- Degrees of freedom: two coordinates (r, θ); steady flow, so no time dependence enters the field.
- θ is multivalued around the origin; where a stream function is used, the range $0 < \theta < 2\pi$ is adopted (cut along the positive x-axis).

Symbol	Meaning	SI unit
Q	source strength (volume flux per unit depth)	$\mathrm{m^2/s}$
U_∞	velocity of the uniform stream, along $+x$	m/s
φ	velocity potential	$\mathrm{m^2/s}$
ψ	stream function	$\mathrm{m^2/s}$
$r_s = Q/(2\pi U_\infty)$	distance source-stagnation point (derived in 3.4)	m

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame at rest with the source. Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y)$ and polar basis $(\mathbf{e}_r, \mathbf{e}_\theta)$ with $x = r\cos\theta, y = r\sin\theta$.
- Change of basis (rotation by θ): $\mathbf{e}_r = \cos\theta\,\mathbf{e}_x + \sin\theta\,\mathbf{e}_y$, $\mathbf{e}_\theta = -\sin\theta\,\mathbf{e}_x + \cos\theta\,\mathbf{e}_y$, so that

$$\begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} v_r \\ v_\theta \end{pmatrix}.$$

This matrix has determinant $\cos^2\theta + \sin^2\theta = 1$ and its inverse is its transpose.

- Gradient in polar coordinates (scale factors $h_r = 1$, $h_\theta = r$, Eq. 2.3): $\nabla\varphi = \partial_r\varphi\,\mathbf{e}_r + r^{-1}\partial_\theta\varphi\,\mathbf{e}_\theta$.
- For part 5, local non-dimensional coordinates around the stagnation point and a non-dimensional time are introduced.

PHASE 3

Step-by-Step Derivation

3.1 Velocity components

Why this tool: for a potential flow $\mathbf{v} = \nabla\varphi$ (Eq. 2.34); in polar coordinates the gradient has the scale factors above.

$$v_r = \frac{\partial\varphi}{\partial r} = \frac{Q}{2\pi r} + U_\infty\cos\theta, \qquad v_\theta = \frac{1}{r}\frac{\partial\varphi}{\partial\theta} = -U_\infty\sin\theta.$$

Cartesian components by the rotation of Phase 2:

$$v_x = v_r\cos\theta - v_\theta\sin\theta = \frac{Q\cos\theta}{2\pi r} + U_\infty\left(\cos^2\theta + \sin^2\theta\right) = \boxed{\frac{Q}{2\pi}\frac{x}{x^2+y^2} + U_\infty,}$$

$$v_y = v_r\sin\theta + v_\theta\cos\theta = \frac{Q\sin\theta}{2\pi r} + U_\infty\left(\sin\theta\cos\theta - \sin\theta\cos\theta\right) = \boxed{\frac{Q}{2\pi}\frac{y}{x^2+y^2}.}$$

Flux check: on any circle of radius r around the source, $\oint v_r\,r\,d\theta = \int_0^{2\pi}\left(\frac{Q}{2\pi} + U_\infty r\cos\theta\right)d\theta = Q + U_\infty r\,[\sin\theta]_0^{2\pi} = Q$, so Q is indeed the source flux.

3.2 Streamlines

Why this tool: the flow is planar and incompressible, $\nabla\cdot\mathbf{v} = r^{-1}\partial_r(rv_r) + r^{-1}\partial_\theta v_\theta = r^{-1}\left(U_\infty\cos\theta\right) - r^{-1}U_\infty\cos\theta = 0$, so the stream lines are the level curves of a stream function (Eq. 2.42). They are obtained directly from the tangency condition (Eq. 2.21) in polar coordinates, $dr/v_r = r\,d\theta/v_\theta$:

$$v_\theta\,dr - r\,v_r\,d\theta = 0 \iff U_\infty \sin\theta\,dr + \left(\frac{Q}{2\pi} + U_\infty r\cos\theta\right) d\theta = 0.$$

This is an exact differential, because

$$d\left[\frac{Q\theta}{2\pi} + U_\infty r\sin\theta\right] = \frac{Q}{2\pi}d\theta + U_\infty\sin\theta\,dr + U_\infty r\cos\theta\,d\theta.$$

Hence, with $\psi \equiv \dfrac{Q\theta}{2\pi} + U_\infty r\sin\theta$,

$$\boxed{\psi = \frac{Q\theta}{2\pi} + U_\infty r\sin\theta = \text{const}, \quad \text{i.e.} \quad \frac{Q\theta}{2\pi} + U_\infty r\sin\theta = \frac{Q\theta_0}{2\pi} + U_\infty r_0\sin\theta_0.}$$

Check in Cartesian variables: $\psi = \frac{Q}{2\pi}\theta + U_\infty y$ gives $\partial_y\psi = \frac{Q}{2\pi}\frac{x}{x^2+y^2} + U_\infty = v_x$ and $-\partial_x\psi = \frac{Q}{2\pi}\frac{y}{x^2+y^2} = v_y$, using $\partial_x\theta = -y/(x^2+y^2)$ and $\partial_y\theta = x/(x^2+y^2)$; this is the sign convention of Eq. (2.41).

3.3 Trajectories and paths

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12). In polar coordinates the system is

$$\frac{dr}{dt} = \frac{Q}{2\pi r} + U_\infty\cos\theta, \qquad r\frac{d\theta}{dt} = -U_\infty\sin\theta, \qquad r(0) = r_i,\ \theta(0) = \theta_i.$$

Because the field is steady, the path lines coincide with the streamlines of 3.2, so the path is

$$r(\theta) = \frac{\dfrac{Q}{2\pi}(\theta_i - \theta) + U_\infty r_i\sin\theta_i}{U_\infty\sin\theta}.$$

The time law follows from the second equation, $dt = -r\,d\theta/(U_\infty\sin\theta)$, by substituting $r(\theta)$ and defining $r_s \equiv Q/(2\pi U_\infty)$:

$$t = -\frac{1}{U_\infty}\int_{\theta_i}^{\theta} \left[r_s(\theta_i - \theta') + r_i \sin\theta_i\right] \frac{d\theta'}{\sin^2\theta'}.$$

Two primitives are needed. First, $\int \csc^2\theta'\,d\theta' = -\cot\theta'$. Second, $F(\theta') \equiv -(\theta_i - \theta')\cot\theta' - \ln|\sin\theta'|$ satisfies $F'(\theta') = \cot\theta' + (\theta_i - \theta')\csc^2\theta' - \cot\theta' = (\theta_i - \theta')\csc^2\theta'$. Barrow's rule between θ_i and θ (where $F(\theta_i) = -\ln|\sin\theta_i|$ because the first term vanishes):

$$\int_{\theta_i}^{\theta} (\theta_i - \theta')\csc^2\theta'\,d\theta' = -(\theta_i - \theta)\cot\theta - \ln\left|\frac{\sin\theta}{\sin\theta_i}\right|, \qquad \int_{\theta_i}^{\theta} \csc^2\theta'\,d\theta' = \cot\theta_i - \cot\theta.$$

Therefore

$$\boxed{t = \frac{1}{U_\infty}\left\{r_s\left[(\theta_i - \theta)\cot\theta + \ln\left|\frac{\sin\theta}{\sin\theta_i}\right|\right] + r_i\sin\theta_i\left(\cot\theta - \cot\theta_i\right)\right\},}$$

valid while the particle stays in the same half-plane ($\sin\theta$ keeps its sign). For $0 < \theta < \pi$, $d\theta/dt < 0$: particles drift towards $\theta \to 0$, i.e. downstream. The axes $\theta = 0, \pi$ are invariant lines, with $dr/dt = Q/(2\pi r) \pm U_\infty$.

3.4 Stagnation points and the streamlines reaching them

Why this tool: a stagnation point is a point where the velocity vanishes (Eq. 2.11), so $v_r = v_\theta = 0$ simultaneously.

$v_\theta = -U_\infty \sin\theta = 0$ gives $\theta = 0$ or $\theta = \pi$. Then $v_r = 0$ requires $\dfrac{Q}{2\pi r} = -U_\infty\cos\theta$. For $\theta = 0$ the right side is negative, impossible for $r > 0$. For $\theta = \pi$,

$$\boxed{\theta = \pi, \quad r = r_s = \frac{Q}{2\pi U_\infty} \quad \Longleftrightarrow \quad (x, y) = (-r_s, 0),}$$

the only stagnation point, upstream of the source.

Streamlines through it: $\psi_s = \dfrac{Q\pi}{2\pi} + U_\infty r_s \sin\pi = \dfrac{Q}{2}$. The streamline $\psi = Q/2$ consists of two branches:

- the incoming axis $\theta = \pi$, $r > r_s$ (where $\sin\pi = 0$ and $\psi = Q/2$);
- the curve $\dfrac{Q\theta}{2\pi} + U_\infty r\sin\theta = \dfrac{Q}{2}$, that is

$$\boxed{r = \frac{Q}{2\pi U_\infty}\,\frac{\pi-\theta}{\sin\theta} = r_s\,\frac{\pi-\theta}{\sin\theta}, \quad 0<\theta<2\pi,}$$

which at $\theta \to \pi$ gives $r \to r_s$ (the limit of $(\pi-\theta)/\sin\theta$ is 1) and, for $\theta \to 0^+$ or $2\pi^-$, $r\sin\theta \to \pm\pi r_s = \pm Q/(2U_\infty)$. The closed-nose body so defined is the Rankine half-body: it has total half-width $Q/(2U_\infty)$ on each side far downstream, hence full width Q/U_∞, consistent with the source flux Q being carried within the width Q/U_∞ of the uniform stream. Fluid emitted by the source stays inside this body, and the incoming stream passes outside.

Sketch (in words): the uniform stream comes from $x \to -\infty$, splits at the stagnation point $S = (-r_s, 0)$, and the dividing streamline leaves S perpendicular to the axis, bending downstream to the two asymptotes $y = \pm Q/(2U_\infty)$.

3.5 Flow near the stagnation point

Why this tool: near a stagnation point the velocity is small, and a first-order Taylor expansion of the field captures the local kinematics (compare Eq. 2.46). Local non-dimensional variables, with $|\xi|, |\eta| \ll 1$ and the time scale r_s/U_∞:

$$\xi = \frac{x + r_s}{r_s}, \qquad \eta = \frac{y}{r_s}, \qquad \tau = \frac{U_\infty t}{r_s}.$$

Use $Q = 2\pi r_s U_\infty$ and $x = r_s(\xi - 1)$ in the Cartesian components of 3.1:

$$\frac{v_x}{U_\infty} = 1 + \frac{\xi - 1}{(\xi-1)^2 + \eta^2}, \qquad \frac{v_y}{U_\infty} = \frac{\eta}{(\xi-1)^2 + \eta^2}.$$

To first order, $\left[(1-\xi)^2 + \eta^2\right]^{-1} = 1 + 2\xi + \mathcal{O}(2)$, so $(\xi - 1)(1 + 2\xi) = -1 - \xi + \mathcal{O}(2)$ and

$$V_\xi \equiv \frac{v_x}{U_\infty} \simeq -\xi, \qquad V_\eta \equiv \frac{v_y}{U_\infty} \simeq \eta.$$

Streamlines: $\dfrac{d\xi}{-\xi} = \dfrac{d\eta}{\eta} \Rightarrow \ln|\xi| + \ln|\eta| = \text{const}$, that is

$$\boxed{\xi\eta = \xi_0\eta_0 \ \ (\text{hyperbolae}).}$$

Trajectories: $\dfrac{d\xi}{d\tau} = -\xi$ and $\dfrac{d\eta}{d\tau} = \eta$, with Barrow's rule, $\ln(\xi/\xi_0) = -\tau$ and $\ln(\eta/\eta_0) = \tau$:

$$\boxed{\xi = \xi_0 e^{-\tau}, \qquad \eta = \eta_0 e^{\tau}.}$$

The product $\xi\eta$ is constant along each trajectory, so paths and streamlines coincide. The axis $\eta = 0$ is the incoming streamline (approached only as $\tau \to \infty$) and $\xi = 0$ is the outgoing streamline along which the body contour leaves the point; the velocity gradient there has eigenvalues $-U_\infty/r_s$ (compression along x) and $+U_\infty/r_s$ (extension along y), a saddle.

Reading of the official handwritten solution

In K4.pdf (part 5), as legible in the scan, the local variable is defined as $\xi = (x - r_0)/r_0$ with $r_0 = Q/(2\pi U_\infty)$. The stagnation point found in part 4 of the same solution is at $\theta = \pi$, i.e. $x = -r_0$, so the local variable must be $\xi = (x + r_0)/r_0$ (written r_s here); otherwise ξ would not be small at the stagnation point. With the corrected definition the handwritten results $V_\xi = -\xi, V_\eta = \eta, \xi\eta = \xi_0\eta_0, \xi = \xi_0 e^{-\tau}, \eta = \eta_0 e^{\tau}$ are recovered. The handwritten solution also reuses r_0 both for the initial radius of a particle (part 3) and for the stagnation distance; here r_i and r_s are used.

PHASE 4

Interpretation, Limits and Dimensional Check

- Far field: $r \to \infty$ gives $\mathbf{v} \to U_\infty \mathbf{e}_x$; near the source $v_r \to Q/(2\pi r)$, a purely radial outflow.

- Limits: $Q \to 0$ gives $r_s \to 0$ and the uniform stream, with streamlines $y = \text{const}$ (3.2 reduces to $r \sin\theta = \text{const}$); $U_\infty \to 0$ gives $r_s \to \infty$, the pure source with radial streamlines $\theta = \text{const}$.
- Mass balance of the half-body: width at infinity Q/U_∞ times speed U_∞ equals Q; this is the asymptotic result $r \sin\theta \to Q/2U_\infty$ of 3.4.
- Consistency: at the nose of the body the streamline direction is perpendicular to the axis (the contour is tangent to the outgoing axis $\xi = 0$ of 3.5), as sketched in the official figure.
- Dimensions (SI): $[Q/2\pi r] = (\mathrm{m}^2/\mathrm{s})/\mathrm{m} = \mathrm{m/s}$; $[r_s] = [Q/U_\infty] = \mathrm{m}$; $[\psi] = [Q] = [U_\infty r] = \mathrm{m}^2/\mathrm{s}$; $[r_s/U_\infty] = \mathrm{s}$, the time scale of 3.5 and of 3.3.
- Numerical cross-check with $Q = 1.7$ m^2/s, $U_\infty = 0.9$ m/s ($r_s = 0.30061$ m), start $r_i = 1.2$ m, $\theta_i = 2.0$ rad: RK4 integration of the particle ODE up to $t = 1.5$ s, then the boxed $t(\theta)$ evaluated at the final angle returns 1.5000000000000147 s, and ψ is conserved (1.5231680274841803 initially, 1.5231680274841746 finally). Asymptote: $r \sin\theta$ at $\theta = 10^{-6}$ rad is 0.944444 m versus $Q/2U_\infty = 0.944444$ m. Linearisation at $\xi = \eta = 10^{-3}$: $v_x/U_\infty = -0.00100000$, $v_y/U_\infty = +0.00100200$.

PROBLEM K5

Velocity potential | Polar coordinates | Acceleration | Unsteady

Oscillating Planar Dipole

Original statement · K5.pdf

OFFICIAL PROBLEM STATEMENT

For the planar fluid motion deriving from the velocity potential

$$\varphi = M\cos(\Omega t)\,\frac{\cos\theta}{r},$$

obtain:

1. Radial and azimuthal velocity components.
2. Expansion rate $\nabla\cdot\vec{v}$.
3. Vorticity.
4. Acceleration of the fluid particle initially located at $r=1$ and $\theta=\pi/2$.
5. Initial value of the circulation around a circle of radius $R=1/2$ centered at $(r=1,\theta=\pi/2)$.
6. Streamlines.
7. Trajectories and particle paths.

Source: K5.pdf (page 1). Notes.pdf, Chapter 2: gradient, divergence and curl in orthogonal coordinates Eqs. (2.3), (2.5), (2.6); potential Eq. (2.34); material derivative and acceleration Eqs. (2.23)-(2.27); circulation and Stokes theorem Eqs. (2.29)-(2.33); trajectories Eq. (2.12); stream lines Eq. (2.21).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar, unsteady, potential flow $\mathbf{v} = \nabla\varphi$ in the plane (r, θ), for $r > 0$. The potential is that of a dipole whose strength $M \cos \Omega t$ oscillates in time.
- Degrees of freedom: two spatial coordinates (r, θ) plus time; the only time scale is Ω^{-1}.
- Dimensions of the constants follow from φ (a velocity times a length, m^2/s): $M \cos \Omega t / r$ has the units of φ, so M has units m^3/s.
- The origin is a singular point and is excluded.

Symbol	Meaning	SI unit
M	dipole strength (per unit depth)	m^3/s
Ω	angular frequency	s^{-1}
r, θ	polar coordinates (the data $r = 1$, $R = 1/2$ are lengths in m)	m, rad
φ	velocity potential	m^2/s

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame at rest with the dipole; polar coordinates (r, θ) with scale factors $(1, r)$, $x = r\cos\theta$, $y = r\sin\theta$, and $\mathbf{e}_z = \mathbf{e}_r \wedge \mathbf{e}_\theta$.
- Gradient: $\nabla\varphi = \partial_r\varphi\, \mathbf{e}_r + r^{-1}\partial_\theta\varphi\, \mathbf{e}_\theta$; divergence: $\nabla \cdot \mathbf{v} = r^{-1}\left[\partial_r(r v_r) + \partial_\theta v_\theta\right]$; vorticity: $\omega_z = r^{-1}\left[\partial_r(r v_\theta) - \partial_\theta v_r\right]$ (Eqs. 2.3, 2.5, 2.6).
- Acceleration in curvilinear coordinates: the component form $a_i = Dv_i/Dt$ of Eq. (2.27) holds only in Cartesian coordinates. The coordinate-free form of Eq. (2.26), $\mathbf{a} = \partial_t\mathbf{v} + \nabla\left(|\mathbf{v}|^2/2\right) - \mathbf{v} \wedge (\nabla \wedge \mathbf{v})$, is used instead.
- For part 7 the material coordinates are the initial position (r_0, θ_0) at $t = 0$.

PHASE 3

Step-by-Step Derivation

3.1 Velocity components

Why this tool: since the motion derives from a potential, $\mathbf{v} = \nabla\varphi$ (Eq. 2.34), with the polar gradient above. Write $c(t) \equiv \cos\Omega t$.

$$v_r = \frac{\partial\varphi}{\partial r} = Mc\cos\theta\,\frac{\partial}{\partial r}\left(\frac{1}{r}\right) = \boxed{-\frac{Mc\cos\theta}{r^2},} \qquad v_\theta = \frac{1}{r}\frac{\partial\varphi}{\partial\theta} = \frac{1}{r}\cdot\frac{Mc}{r}\,(-\sin\theta) = \boxed{-\frac{Mc\sin\theta}{r^2}.}$$

3.2 Expansion rate

Why this tool: $\nabla\cdot\mathbf{v}$ is the rate of volume change per unit volume (Eqs. 2.62-2.63).

$$r\,v_r = -\frac{Mc\cos\theta}{r} \;\Rightarrow\; \frac{\partial(rv_r)}{\partial r} = \frac{Mc\cos\theta}{r^2}, \qquad \frac{\partial v_\theta}{\partial\theta} = -\frac{Mc\cos\theta}{r^2},$$

$$\nabla\cdot\mathbf{v} = \frac{1}{r}\left[\frac{Mc\cos\theta}{r^2} - \frac{Mc\cos\theta}{r^2}\right] = \boxed{0.}$$

The flow is incompressible for all t (consistent with $\nabla^2\varphi = 0$).

3.3 Vorticity

Why this tool: $\omega_z = r^{-1}\left[\partial_r(rv_\theta) - \partial_\theta v_r\right]$ is twice the angular velocity of the fluid element (Eq. 2.52).

$$r\,v_\theta = -\frac{Mc\sin\theta}{r} \;\Rightarrow\; \frac{\partial(rv_\theta)}{\partial r} = \frac{Mc\sin\theta}{r^2}, \qquad \frac{\partial v_r}{\partial\theta} = \frac{Mc\sin\theta}{r^2},$$

$$\omega_z = \frac{1}{r}\left[\frac{Mc\sin\theta}{r^2} - \frac{Mc\sin\theta}{r^2}\right] = \boxed{0.}$$

The motion is irrotational, as it must be for $\mathbf{v} = \nabla\varphi$ ($\nabla \wedge \nabla\varphi \equiv 0$).

3.4 Acceleration of the particle initially at $r = 1, \theta = \pi/2$

Why this tool: the acceleration is the material derivative of the velocity (Eq. 2.25). Because $\boldsymbol{\omega} = 0$, Eq. (2.26) reduces to $\mathbf{a} = \partial_t \mathbf{v} + \nabla(|\mathbf{v}|^2/2)$, which is valid in polar coordinates.

Local term, using $\partial_t c = -\Omega\sin\Omega t$:

$$\frac{\partial v_r}{\partial t} = \frac{M\Omega\sin\Omega t\cos\theta}{r^2}, \qquad \frac{\partial v_\theta}{\partial t} = \frac{M\Omega\sin\Omega t\sin\theta}{r^2}.$$

Convective term: $|\mathbf{v}|^2 = v_r^2 + v_\theta^2 = \dfrac{M^2c^2\left(\cos^2\theta + \sin^2\theta\right)}{r^4} = \dfrac{M^2c^2}{r^4}$, hence

$$\frac{|\mathbf{v}|^2}{2} = \frac{M^2c^2}{2r^4}, \qquad \frac{\partial}{\partial r}\left(\frac{|\mathbf{v}|^2}{2}\right) = -\frac{2M^2c^2}{r^5}, \qquad \frac{1}{r}\frac{\partial}{\partial\theta}\left(\frac{|\mathbf{v}|^2}{2}\right) = 0.$$

Field acceleration:

$$a_r = \frac{M\Omega\sin\Omega t\cos\theta}{r^2} - \frac{2M^2\cos^2\Omega t}{r^5}, \qquad a_\theta = \frac{M\Omega\sin\Omega t\sin\theta}{r^2}.$$

Evaluate at $r = 1, \theta = \pi/2$ ($\cos\theta = 0, \sin\theta = 1$):

$$\boxed{\mathbf{a}(1, \tfrac{\pi}{2}, t) = -2M^2\cos^2(\Omega t)\,\mathbf{e}_r + M\Omega\sin(\Omega t)\,\mathbf{e}_\theta.}$$

This is the acceleration of whichever particle occupies the point $(1, \pi/2)$ at time t. For the particle that is at that point at the initial instant, set $t = 0$: $\mathbf{a} = -2M^2\mathbf{e}_r$ (the local term vanishes because $\sin 0 = 0$). For $t > 0$ that particle has moved away from the point, so its

acceleration is the field value at its new position (part 7).

3.5 Initial circulation around the circle of radius $1/2$ centred at $(r = 1, \theta = \pi/2)$

Why this tool: $\Gamma = \oint \mathbf{v} \cdot d\mathbf{l}$ (Eq. 2.29) equals the flux of the vorticity through any surface bounded by the curve that lies in the fluid (Eq. 2.32). The centre is at Cartesian $(0, 1)$ and the circle has radius $1/2$, so every point of the disc is at a distance of at least $1/2$ from the origin: the disc lies entirely inside the domain, where $\omega_z = 0$.

$$\Gamma = \iint_S \omega_z \, d\sigma = \iint_S 0 \, d\sigma = \boxed{0}$$

at $t = 0$ and, since $\omega_z = 0$ for all t, at every instant. In fact φ is single-valued, so $\oint \nabla\varphi \cdot d\mathbf{l} = \varphi(\text{end}) - \varphi(\text{start}) = 0$ for every closed curve, including those that enclose the origin.

3.6 Streamlines

Why this tool: streamlines are tangent to $\mathbf{v}$ at a frozen instant (Eq. 2.21): $dr/v_r = r\,d\theta/v_\theta$. At an instant with $\cos\Omega t \neq 0$ the common factor $-Mc/r^2$ cancels:

$$\frac{dr}{\cos\theta} = \frac{r\,d\theta}{\sin\theta} \implies \frac{dr}{r} = \frac{\cos\theta}{\sin\theta}\,d\theta \implies \ln\frac{r}{r_0} = \ln\left|\frac{\sin\theta}{\sin\theta_0}\right|,$$

$$\boxed{r = \frac{r_0}{\sin\theta_0}\sin\theta = C\sin\theta.}$$

In Cartesian variables $r^2 = Cr\sin\theta$ gives $x^2 + y^2 = Cy$: circles of diameter C centred at $(0, C/2)$ and tangent to the x-axis at the origin. The shape does not depend on t. (At the isolated instants with $\cos\Omega t = 0$ the whole velocity field vanishes.) Equivalent form: the stream function $\psi = -Mc\sin\theta/r$ (with $v_r = r^{-1}\partial_\theta\psi$, $v_\theta = -\partial_r\psi$) has the level curves $\sin\theta/r = \text{const}$.

3.7 Trajectories and particle paths

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12), here

$$\frac{dr}{dt} = -\frac{Mc\cos\theta}{r^2}, \qquad r\frac{d\theta}{dt} = -\frac{Mc\sin\theta}{r^2}, \qquad r(0) = r_0,\ \theta(0) = \theta_0.$$

Dividing the first equation by the second eliminates both t and the factor $c(t)$:

$$\frac{dr}{r\,d\theta} = \frac{\cos\theta}{\sin\theta} \implies \boxed{r = r_0\frac{\sin\theta}{\sin\theta_0}}$$

which is the path line. It coincides with the streamlines of 3.6 because the direction of $\mathbf{v}$ does not depend on time (only its magnitude does).

Time law. Substitute the path into the second equation, $r^3 = r_0^3\sin^3\theta/\sin^3\theta_0$:

$$\frac{d\theta}{dt} = -\frac{Mc\sin\theta}{r^3} = -\frac{M\sin^3\theta_0}{r_0^3}\,\frac{\cos\Omega t}{\sin^2\theta} \implies \sin^2\theta\,d\theta = -\frac{M\sin^3\theta_0}{r_0^3}\cos\Omega t\,dt.$$

Integrate with Barrow's rule between $(\theta_0, 0)$ and (θ, t). For the left side use $\sin^2\theta' = (1-\cos 2\theta')/2$, whose primitive is $\theta'/2 - \sin 2\theta'/4$; for the right side, $\int_0^t \cos\Omega t'\,dt' = \sin(\Omega t)/\Omega$:

$$\boxed{\frac{\theta-\theta_0}{2} - \frac{\sin 2\theta - \sin 2\theta_0}{4} = -\frac{M\sin^3\theta_0}{r_0^3\,\Omega}\sin\Omega t.}$$

Together with $r = r_0\sin\theta/\sin\theta_0$ this defines $\mathbf{x}(t)$ implicitly. The right-hand side vanishes whenever $\sin\Omega t = 0$, so the particle returns to its starting point every half period π/Ω: it oscillates back and forth along its circular path. (Particles on the axis $\theta_0 = 0$ or π stay on it, with $r^3 = r_0^3 \mp 3(M/\Omega)\sin\Omega t$.)

PHASE 4

Interpretation, Limits and Dimensional Check

- The flow is the oscillating field of a point dipole: incompressible, irrotational, with streamlines that are circles through the singular point and a velocity that decays as r^{-2}.
- Steady limit $\Omega \to 0$ (with $c \to 1$): the local term vanishes and the acceleration is purely convective, $a_r = -2M^2/r^5$ and $a_\theta = 0$, equal to $\nabla(|\mathbf{v}|^2/2)$; the particle then drifts monotonically along the circle.
- Consistency of the answers: part 5 ($\Gamma = 0$) follows from part 3 ($\omega_z = 0$); part 7 reproduces part 6 geometrically.
- Dimensions (SI): $[v] = [M/r^2] = (\mathrm{m}^3/\mathrm{s})/\mathrm{m}^2 = \mathrm{m/s}$; $[a] = [M^2/r^5] = (\mathrm{m}^6/\mathrm{s}^2)/\mathrm{m}^5 = \mathrm{m/s}^2$ and $[M\Omega/r^2] = (\mathrm{m}^3/\mathrm{s})(1/\mathrm{s})/\mathrm{m}^2 = \mathrm{m/s}^2$; the combination $M/(r_0^3\Omega)$ in 3.7 is $(\mathrm{m}^3/\mathrm{s})/(\mathrm{m}^3/\mathrm{s}) = 1$, dimensionless as required for an angle.
- Numerical cross-check: symbolic differentiation of φ gives $\nabla \cdot \mathbf{v} = 0$, $\omega_z = 0$ and, from the full material derivative in polar components (including the $-v_\theta^2/r$ and $v_r v_\theta/r$ terms), the same a_r, a_θ as above, in particular $\mathbf{a}(1, \pi/2, t) = -2M^2 \cos^2 \Omega t\, \mathbf{e}_r + M\Omega \sin \Omega t\, \mathbf{e}_\theta$. RK4 integration of the particle ODE with $M = 0.8$ m^3/s, $\Omega = 1.7$ s^{-1}, $r_0 = 1$ m, $\theta_0 = 0.45\pi$ up to $t = 1.3$ s gives $r = 0.846496$ m, equal to $r_0 \sin\theta / \sin\theta_0$ at the integrated angle, and the two sides of the boxed time law, -0.363902 and -0.363902.

> **READING OF THE OFFICIAL HANDWRITTEN SOLUTION**
>
> K5.pdf evaluates the acceleration at the field point $(1, \pi/2)$ for general t, obtaining the vector given in 3.4; the value for the particle that is there at $t = 0$ is its $t = 0$ value. The handwritten trajectory law agrees with the boxed relation in 3.7 and notes that streamlines and path lines coincide. No numerical discrepancy was found.

PROBLEM K6

Axisymmetric | Incompressibility | Spiral path lines

Three-Dimensional Burgers Vortex

Original statement · K6.pdf

OFFICIAL PROBLEM STATEMENT

The axisymmetric velocity field corresponding to a three-dimensional steady vortex of circulation Γ is given in cylindrical coordinates by

$$v_r = -\frac{r}{\tau_r}, \qquad v_\theta = \frac{\Gamma}{2\pi r}\left\{1 - \exp\left[-\left(\frac{r}{R_o}\right)^2\right]\right\}, \qquad v_z = \frac{z}{\tau_z},$$

where the constants R_o, τ_r and τ_z measure the radius of the inner core where viscous effects are important, and the characteristic times associated with radial convection and axial stretch, respectively.

1. Determine the condition required for the motion to be that of an incompressible fluid.
2. Compute streamlines and trajectories.

Source: K6.pdf (page 1). Notes.pdf, Chapter 2: divergence in orthogonal coordinates Eq. (2.5) with the cylindrical scale factors $(1, r, 1)$; incompressibility Eq. (2.38) and (2.63); trajectories Eqs. (2.12)-(2.13); stream lines Eq. (2.21).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Steady, axisymmetric, three-dimensional flow with three velocity components (v_r, v_θ, v_z) that depend on r and z only. A particle has three degrees of freedom (r, θ, z) .
- R_o, τ_r and τ_z are positive constants: $v_r < 0$ is an inflow (radial convection towards the axis) and v_z has the sign of z (axial stretching away from the plane $z = 0$).
- The vortex has circulation Γ far from the core, $r v_\theta \to \Gamma / 2\pi$; the Gaussian factor regularises the axis.
- The velocity is smooth at $r = 0$: expanding the exponential, $v_\theta \simeq \Gamma r / (2\pi R_o^2)$.

Symbol	Meaning	SI unit
Γ	circulation of the vortex	$\mathrm{m^2/s}$
R_o	core radius	m
τ_r	time of radial convection	s
τ_z	time of axial stretching	s
r, z	radial and axial coordinates	m

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame with the vortex axis along z; cylindrical coordinates (r, θ, z) with scale factors $(1, r, 1)$ and basis $(\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z)$.
- Divergence (Eq. 2.5 with $h_1 = 1, h_2 = r, h_3 = 1$): $\nabla \cdot \mathbf{v} = \frac{1}{r}\left[\frac{\partial (r v_r)}{\partial r} + \frac{\partial v_\theta}{\partial \theta} + \frac{\partial (r v_z)}{\partial z}\right]$.
- Lagrangian labels: (r_0, θ_0, z_0) at $t = 0$. The change of variable $t \to r$ along a trajectory, $dt = -\tau_r \, dr/r$ (from the radial equation), will be used to integrate the azimuthal motion; the substitution $u = r^2$ is then used for the primitive.
- The trajectory system is not coupled in the cylindrical basis for r and z (each depends only on its own coordinate), and θ follows from $r(t)$.

PHASE 3

Step-by-Step Derivation

3.1 Condition for incompressibility

Why this tool: for a liquid (constant density) the volume of a fluid element cannot change, which requires $\nabla \cdot \mathbf{v} = 0$ at every point (Eqs. 2.38 and 2.63).

$$\nabla \cdot \mathbf{v} = \frac{1}{r}\left[\frac{\partial}{\partial r}\left(-\frac{r^2}{\tau_r}\right) + \frac{\partial v_\theta}{\partial \theta} + r\frac{\partial}{\partial z}\left(\frac{z}{\tau_z}\right)\right] = \frac{1}{r}\left[-\frac{2r}{\tau_r} + 0 + \frac{r}{\tau_z}\right] = -\frac{2}{\tau_r} + \frac{1}{\tau_z},$$

where $\partial_\theta v_\theta = 0$ because v_θ does not depend on θ (axisymmetry). The expansion rate is the same constant everywhere, and it vanishes if and only if

$$\boxed{\tau_r = 2\tau_z.}$$

The radial inflow convergence $(2/\tau_r)$ must be exactly balanced by the axial stretching $(1/\tau_z)$.

3.2 Trajectories

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12) with $(r,\theta,z)(0) = (r_0,\theta_0,z_0)$. In cylindrical coordinates the components are $dr/dt = v_r$, $r\,d\theta/dt = v_\theta$, $dz/dt = v_z$.

Radial motion:

$$\frac{dr}{dt} = -\frac{r}{\tau_r} \implies \int_{r_0}^{r} \frac{dr'}{r'} = -\frac{1}{\tau_r}\int_0^t dt' \implies \ln\frac{r}{r_0} = -\frac{t}{\tau_r} \implies \boxed{r = r_0\, e^{-t/\tau_r}.}$$

Axial motion:

$$\frac{dz}{dt} = \frac{z}{\tau_z} \implies \ln\frac{z}{z_0} = \frac{t}{\tau_z} \implies \boxed{z = z_0\, e^{t/\tau_z}.}$$

Azimuthal motion:

$$\frac{d\theta}{dt} = \frac{\Gamma}{2\pi r^2}\left[1 - e^{-(r/R_o)^2}\right].$$

Change of variable from t to r using $dt = -\tau_r\, dr/r$:

$$d\theta = -\frac{\Gamma\tau_r}{2\pi}\,\frac{1 - e^{-(r/R_o)^2}}{r^3}\,dr \implies \theta - \theta_0 = -\frac{\Gamma\tau_r}{2\pi}\int_{r_0}^{r} \frac{1 - e^{-(s/R_o)^2}}{s^3}\,ds.$$

Substitute $u = s^2$, $du = 2s\,ds$, so $ds/s^3 = du/(2u^2)$, and write $a = 1/R_o^2$:

$$\theta - \theta_0 = -\frac{\Gamma\tau_r}{4\pi}\int_{r_0^2}^{r^2} \frac{1 - e^{-au}}{u^2}\,du.$$

A primitive is $G(u) = -\dfrac{1 - e^{-au}}{u} - a\,E_1(au)$, where $E_1(x) = \int_x^\infty e^{-s}\,ds/s$ is the exponential integral, $E_1'(x) = -e^{-x}/x$. Check by differentiation:

$$G'(u) = \frac{1-e^{-au}}{u^2} - \frac{a\,e^{-au}}{u} + a\cdot\frac{e^{-au}}{u} = \frac{1-e^{-au}}{u^2}.$$

Barrow's rule between $u = r_0^2$ and $u = r^2$ gives

$$\boxed{\theta - \theta_0 = -\frac{\Gamma\tau_r}{4\pi}\left[G(r^2) - G(r_0^2)\right], \qquad r = r_0 e^{-t/\tau_r},}$$

which gives $\theta(t)$ explicitly once $r(t)$ is inserted. Two limits of the integral: for a particle inside the core ($r_0, r \ll R_o$), $1 - e^{-s^2/R_o^2} \simeq s^2/R_o^2$ and $\theta - \theta_0 \simeq -\frac{\Gamma\tau_r}{2\pi R_o^2}\ln\frac{r}{r_0} = \frac{\Gamma}{2\pi R_o^2}\,t$ (solid-body rotation with angular velocity $\Gamma/2\pi R_o^2$); for $r_0, r \gg R_o$, the bracket tends to 1 and $\theta - \theta_0 \simeq \frac{\Gamma\tau_r}{4\pi}\left(\frac{1}{r^2} - \frac{1}{r_0^2}\right)$, i.e. the angle advances ever faster as the particle is convected inwards (conservation of $r v_\theta$).

Path lines: eliminate t. The relation $\theta(r)$ above is already time-free. For the axial coordinate, $t = -\tau_r \ln(r/r_0)$ and $z = z_0 e^{t/\tau_z} = z_0 (r/r_0)^{-\tau_r/\tau_z}$; with the incompressibility condition $\tau_r/\tau_z = 2$:

$$\boxed{z\,r^2 = z_0\,r_0^2.}$$

3.3 Streamlines

Why this tool: the flow is steady, so streamlines and path lines coincide as curves; they are still obtained from the tangency condition (Eq. 2.21 with scale factors $1, r, 1$):

$$\frac{dr}{v_r} = \frac{r\,d\theta}{v_\theta} = \frac{dz}{v_z}.$$

First equality with the third: $\dfrac{dr}{-r/\tau_r} = \dfrac{dz}{z/\tau_z} \Rightarrow \dfrac{dz}{z} = -\dfrac{\tau_r}{\tau_z}\dfrac{dr}{r} \Rightarrow \ln\dfrac{z}{z_0} = -\dfrac{\tau_r}{\tau_z}\ln\dfrac{r}{r_0}$, i.e. $(z/z_0)^{\tau_z} = (r/r_0)^{-\tau_r}$, which with $\tau_r = 2\tau_z$ is $z\,r^2 = z_0\,r_0^2$. First equality alone: $d\theta = \dfrac{v_\theta}{r\,v_r}\,dr = -\dfrac{\Gamma\tau_r}{2\pi}\dfrac{1-e^{-(r/R_o)^2}}{r^3}\,dr$, the same relation as in 3.2. Hence

$$\boxed{\theta - \theta_0 = -\frac{\Gamma \tau_r}{4\pi}\left[G(r^2) - G(r_0^2)\right], \qquad z\, r^2 = z_0\, r_0^2}$$

are the streamlines: spirals winding around the axis on the surfaces $zr^2 = \text{const}$, tightening as they approach the axis. They are traversed with the time law $r = r_0 e^{-t/\tau_r}$.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: a vortex of circulation Γ whose core is sustained against viscous diffusion by the radial inflow and the axial strain; vortex lines are stretched along z while the core shrinks. For $r \gg R_o$ the flow is the irrotational vortex $v_\theta \to \Gamma/2\pi r$, and for $r \ll R_o$ it is a rigid rotation with vorticity $\omega_z = \Gamma/(\pi R_o^2)$. In general $\omega_z = r^{-1}\partial_r(r v_\theta) = \dfrac{\Gamma}{\pi R_o^2} e^{-r^2/R_o^2}$, a Gaussian vorticity distribution whose integral over the plane is Γ.
- Volume check: a ring of fluid with radius r and height z conserves $r^2 z$ (3.2), as required by $\nabla \cdot \mathbf{v} = 0$. For general τ_r and τ_z the product evolves as $r^2 z = r_0^2 z_0 \exp\left[(1/\tau_z - 2/\tau_r)\, t\right] = r_0^2 z_0 \exp\left[(\nabla \cdot \mathbf{v})\, t\right]$, which is constant exactly when $\tau_r = 2\tau_z$.
- Axis and plane: $r = 0$ and $z = 0$ are invariant (a particle on the axis only moves along it; a particle in the plane $z = 0$ stays in it).
- Dimensions (SI): $[r/\tau_r] = \text{m/s}$; $[\Gamma/2\pi r] = (\text{m}^2/\text{s})/\text{m} = \text{m/s}$; $[z/\tau_z] = \text{m/s}$; $[\nabla \cdot \mathbf{v}] = [1/\tau] = \text{s}^{-1}$; in the integral for θ, $[\Gamma\tau_r/(4\pi)] \cdot [1/u] = (\text{m}^2/\text{s})(\text{s})(1/\text{m}^2)$ is dimensionless, as an angle must be; $a = 1/R_o^2$ has units m^{-2} so au is dimensionless.
- Numerical cross-check with $\Gamma = 1.4\ \text{m}^2/\text{s}$, $R_o = 0.8\ \text{m}$, $\tau_r = 1.3\ \text{s}$, $\tau_z = 0.65\ \text{s}$, initial point $(r_0, \theta_0, z_0) = (1.1\ \text{m}, 0.3\ \text{rad}, 0.7\ \text{m})$, $t = 0.9\ \text{s}$: RK4 integration of the particle ODE gives $r = 0.550462\ \text{m}$ (formula $r_0 e^{-t/\tau_r}$: same), $zr^2 = 0.847000\ \text{m}^3$ (same as $z_0 r_0^2$), and $\theta - \theta_0 = 0.2000243\ \text{rad}$, which is also what the E_1 expression and a direct quadrature of the integral in 3.2 return.

READING OF THE OFFICIAL HANDWRITTEN SOLUTION

K6.pdf obtains $\tau_r = 2\tau_z$, the radial and axial exponentials, the integral form for $\theta - \theta_0$ (left unevaluated) and the streamline relation $zr^2 = z_0 r_0^2$; all agree with the above. Evaluating the integral in terms of the exponential integral E_1, and the core and far-

field limits of $\theta(t)$, are added here.

PROBLEM K7

Potential | Stream function | Strain rate | Fluid element

Hyperbolic Stagnation-Point Flow

Original statement · K7.pdf

OFFICIAL PROBLEM STATEMENT

Consider the planar motion of velocity components $v_x = ax$ and $v_y = by$, where a and b are known constants.

1. If the flow is irrotational compute the associated velocity potential.
2. Compute the value of b for which the flow corresponds to that of an incompressible fluid (use that value in the solution of the following questions).
3. Obtain the stream lines and represent the result schematically.
4. Determine the trajectory and the path of a fluid particle initially located at $x_o = 0$ and $y_o = 1$. How long does it take to reach the origin?
5. Calculate the acceleration of fluid particles located along the axis $y = 0$.
6. Find the stream function for the flow.
7. Compute the rate-of-strain tensor.
8. Investigate the evolution of a fluid element whose initial shape is that of a square of sides dl,

Source: K7.pdf (page 1). The statement of part 8 is cut off in the scan after "sides dl,"; the handwritten solution takes the sides parallel to the coordinate axes, which is assumed here. Notes.pdf, Chapter 2: potential Eq. (2.34), incompressibility Eq. (2.38), stream lines Eq. (2.21), acceleration Eqs. (2.25)-(2.27), stream function Eqs. (2.40)-(2.42), rate-of-strain tensor Eqs. (2.47)-(2.48), square element Eqs. (2.56)-(2.61).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Steady, planar, linear velocity field $\mathbf{v} = (ax, by)$ on the whole plane. Two spatial degrees of freedom; no time dependence.
- a and b are constants with units s^{-1}. For the sketches and for part 4 we take $a > 0$ (a particle on the y-axis moves towards the origin once $b = -a$); for $a < 0$ every velocity reverses.
- The only stagnation point of the field is the origin (where $\mathbf{v} = 0$, Eq. 2.11).

Symbol	Meaning	SI unit
a, b	strain rates along x and y	s^{-1}
$(x_o, y_o) = (0, 1)$	initial position of the particle of part 4	m
dl	side of the fluid element	m

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y)$, origin at the stagnation point. The velocity gradient convention is $(\nabla \mathbf{v})_{ij} = \partial v_j / \partial x_i$.
- Cartesian acceleration: $a_i = \partial_t v_i + v_j \, \partial_j v_i$ (Eq. 2.27, valid in Cartesian coordinates).
- No change of coordinates is required; the principal axes of strain coincide with the coordinate axes (part 7), so the change-of-basis matrix is the identity.

PHASE 3

Step-by-Step Derivation

3.1 Velocity potential for the irrotational case

Why this tool: a velocity potential exists when $\nabla \wedge \mathbf{v} = 0$ (Eq. 2.34).

$$\omega_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = \frac{\partial (by)}{\partial x} - \frac{\partial (ax)}{\partial y} = 0 - 0 = 0$$

for any a and b, so the field is irrotational and $\mathbf{v} = \nabla \varphi$. Integrate $\partial_x \varphi = ax$: $\varphi = ax^2/2 + f(y)$. Then $\partial_y \varphi = f'(y) = by$ gives $f = by^2/2 + \varphi_0$:

$$\boxed{\varphi = \frac{ax^2 + by^2}{2} + \varphi_0.}$$

3.2 Incompressibility

Why this tool: $\nabla \cdot \mathbf{v} = 0$ (Eq. 2.38) is the condition for a liquid element to keep its volume.

$$\nabla \cdot \mathbf{v} = \frac{\partial (ax)}{\partial x} + \frac{\partial (by)}{\partial y} = a + b = 0 \implies \boxed{b = -a.}$$

From here on $\mathbf{v} = (ax, -ay)$ and $\varphi = a(x^2 - y^2)/2$ (up to a constant).

3.3 Streamlines

Why this tool: the field is steady, so the tangency condition (Eq. 2.21) gives fixed curves:

$$\frac{dx}{v_x} = \frac{dy}{v_y} \implies \frac{dx}{ax} = \frac{dy}{-ay} \implies \frac{dx}{x} = -\frac{dy}{y}.$$

Integrate between (x_0, y_0) and (x, y) (for $x_0 y_0 \neq 0$): $\ln|x/x_0| = -\ln|y/y_0|$, so $\ln|xy/(x_0 y_0)| = 0$:

$$\boxed{xy = x_0 y_0 = \text{const.}}$$

These are rectangular hyperbolae with the coordinate axes as asymptotes. The axes themselves are also streamlines ($x_0 y_0 = 0$): the y-axis ($x = 0$), along which $v_y = -ay$ points towards the origin for $a > 0$, and the x-axis ($y = 0$), along which $v_x = ax$ points away from it. Schematic (in words): fluid approaches the origin along the y-axis, turns, and leaves along the x-axis; in each quadrant the streamlines are hyperbolae that follow the incoming axis and then the outgoing axis. This is the flow near a stagnation point on a wall (with the wall along the x-axis, in the half-plane $y > 0$).

3.4 Trajectory and path of the particle starting at $(0, 1)$

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12).

$$\frac{dx}{dt} = ax, \quad x(0) = 0 \implies x(t) = 0; \qquad \frac{dy}{dt} = -ay, \quad y(0) = 1\ \text{m}.$$

Separate variables and apply Barrow's rule:

$$\int_1^y \frac{dy'}{y'} = -a \int_0^t dt' \implies \ln y = -at \implies \boxed{x(t) = 0, \quad y(t) = e^{-at}\ (\text{m}).}$$

Path line: the segment of the y-axis $x = 0, 0 < y \leq 1$ m.

Time to reach the origin. $y = 0$ would require $e^{-at} = 0$, which only happens as $t \to \infty$. The particle approaches the stagnation point exponentially but never reaches it in finite time: it takes the time $t_\varepsilon = a^{-1} \ln(1\,\text{m}/\varepsilon)$ to come within a distance ε of the origin, and $t_\varepsilon \to \infty$ as $\varepsilon \to 0$. (A solution of the ODE cannot reach in finite time an equilibrium point of a Lipschitz field.)

3.5 Acceleration along $y = 0$

Why this tool: the acceleration is the material derivative of the velocity (Eq. 2.25); in Cartesian coordinates $a_i = \partial_t v_i + v_j \partial_j v_i$ (Eq. 2.27). The flow is steady, so only the convective part remains.

$$a_x = v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = (ax)(a) + (-ay)(0) = a^2 x, \qquad a_y = v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} = (ax)(0) + (-ay)(-a) = a^2 y.$$

On the axis $y = 0$:

$$\boxed{\mathbf{a} = a^2 x \, \mathbf{e}_x .}$$

It is directed along $+x$ for $x > 0$ and along $-x$ for $x < 0$, i.e. away from the origin, so the particles on the axis accelerate as they move outwards. Since the flow is irrotational, $\mathbf{a} = \nabla(|\mathbf{v}|^2/2) = a^2(x, y)$ (Eq. 2.26), which agrees.

3.6 Stream function

Why this tool: the flow is planar and incompressible ($b = -a$), so $v_x = \partial_y \psi$ and $v_y = -\partial_x \psi$ (Eq. 2.41).

$$\frac{\partial \psi}{\partial y} = ax \;\Rightarrow\; \psi = axy + f(x); \qquad -\frac{\partial \psi}{\partial x} = -ay - f'(x) = v_y = -ay \;\Rightarrow\; f' = 0.$$

$$\boxed{\psi = axy + \psi_0 .}$$

The level curves $\psi =$ const are the hyperbolae $xy =$ const of 3.3 (Eq. 2.42), and $\psi - \psi_0$ vanishes on both axes.

3.7 Rate-of-strain tensor

Why this tool: $\bar{\bar{T}}_d = \frac{1}{2}\left[\nabla\mathbf{v} + (\nabla\mathbf{v})^T\right]$ (Eq. 2.47) collects the stretching and shearing of material elements.

$$\nabla\mathbf{v} = \begin{pmatrix} \partial_x v_x & \partial_x v_y \\ \partial_y v_x & \partial_y v_y \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix}, \qquad \boxed{\bar{\bar{T}}_d = \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} = \nabla\mathbf{v},}$$

$$\bar{\bar{T}}_r = \frac{1}{2}\left[\nabla\mathbf{v} - (\nabla\mathbf{v})^T\right] = 0 \quad (\text{consistent with } \omega_z = 0).$$

$\bar{\bar{T}}_d$ is already diagonal: the principal directions are the coordinate axes, with principal strain rates $+a$ (along x, extension) and $-a$ (along y, compression); their sum is $\nabla\cdot\mathbf{v} = 0$.

3.8 Evolution of a square fluid element

Why this tool: a material segment evolves as $d\mathbf{l}(t+dt) = d\mathbf{l} + (d\mathbf{l}\cdot\nabla\mathbf{v})\,dt$ (Eqs. 2.46 and 2.56-2.57). With the sides along the axes:

$$d\mathbf{l}_1 = dl\,(1,0) \;\rightarrow\; dl\,(1,0) + dl\,(1,0)\begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} dt = dl\,(1 + a\,dt,\; 0),$$

$$d\mathbf{l}_2 = dl\,(0,1) \;\rightarrow\; dl\,(0,1) + dl\,(0,1)\begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} dt = dl\,(0,\; 1 - a\,dt).$$

The square becomes a rectangle: it is stretched along x and compressed along y at the rate a, with no rotation ($\bar{\bar{T}}_r = 0$) and no shear (off-diagonal terms zero, the right angle is preserved). Its area is $dl^2(1+a\,dt)(1-a\,dt) = dl^2(1-a^2dt^2) \simeq dl^2$ to first order, as required by $\nabla\cdot\mathbf{v} = 0$. For a finite time the same linear flow gives sides $dl\,e^{at}$ and $dl\,e^{-at}$ (from $dl_x/dt = a\,dl_x$, $dl_y/dt = -a\,dl_y$), whose product is exactly dl^2.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: the origin is a saddle-type stagnation point (a stagnation-point flow). The flow compresses material along y and stretches it along x, with $T_{d,xx} + T_{d,yy} = 0$; there is no rotation anywhere.
- Consistency of results: the streamlines $xy = \text{const}$ (3.3) are the level curves of ψ (3.6); the velocity is the gradient of φ (3.1) and perpendicular to its level curves $x^2 - y^2 = \text{const}$, which are the orthogonal family of hyperbolae; $\bar{\bar{T}}_r = 0$ (3.7) agrees with $\omega_z = 0$ (3.1).
- Limits: $a \to 0$ gives a fluid at rest (all quantities vanish, the particle of 3.4 stays at $y = 1$ m); $t \to \infty$ gives $y \to 0$.
- Dimensions (SI): $[a] = \mathrm{s}^{-1}$; $[\varphi] = [ax^2] = \mathrm{m}^2/\mathrm{s}$; $[\psi] = [axy] = \mathrm{m}^2/\mathrm{s}$; $[\mathbf{a}] = [a^2 x] = \mathrm{s}^{-2} \cdot \mathrm{m} = \mathrm{m}/\mathrm{s}^2$; $[\bar{\bar{T}}_d] = \mathrm{s}^{-1}$; $[at]$ is dimensionless.
- Numerical cross-check: the acceleration of 3.5 was recomputed symbolically from $\mathbf{a} = (\mathbf{v} \cdot \nabla)\mathbf{v}$, and $\partial_y \psi$, $-\partial_x \psi$ return the velocity components.

READING OF THE OFFICIAL HANDWRITTEN SOLUTION

K7.pdf obtains $b = -a$, $xy = x_0 y_0$, $x = 0$, $y = e^{-at}$ with "$y \to 0$ when $t \to \infty$" (so the origin is not reached in finite time), $\mathbf{a} = a^2 x\, \mathbf{e}_x$ on $y = 0$, $\psi = axy + \psi_0$, $\bar{\bar{T}}_d = \mathrm{diag}(a, -a)$ and the element $dl\,(1 + a\,dt, 0)$, $dl\,(0, 1 - a\,dt)$. All agree with the results above; no discrepancy was found.

PROBLEM K8

Potential flow | Fluid surface | Path lines | Unsteady

Pulsating Three-Dimensional Potential Flow

Original statement · K8.pdf

OFFICIAL PROBLEM STATEMENT

The velocity potential of a three-dimensional flow field is given by

$$\varphi = A\cos(\Omega t)\left(2x^2 - y^2 - z^2\right),$$

where A and Ω are known constants.

1. Determine the velocity field, verifying that $\nabla \wedge \vec{v} = 0$ and $\nabla \cdot \vec{v} = 0$.
2. Compute the streamlines, trajectories and particle paths.
3. Obtain the fluid surface that at the initial instant is a sphere of radius R centered at $(x, y, z) = (-R/2, 0, 0)$.

Source: K8.pdf (page 1). Notes.pdf, Chapter 2: potential Eq. (2.34); expansion rate Eqs. (2.37)-(2.38); trajectories and path lines Eqs. (2.12)-(2.15); fluid surfaces Eqs. (2.18)-(2.20); stream lines Eq. (2.21).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Three-dimensional, unsteady, potential flow, $\mathbf{v} = \nabla\varphi$, in all of space. The only time dependence is the common factor $\cos\Omega t$.
- Degrees of freedom: three coordinates (x, y, z); the field is linear in them.
- Units: φ is in m^2/s and multiplies a length squared, so A has units s^{-1}.
- At the instants when $\cos\Omega t = 0$ the velocity vanishes everywhere; streamlines are defined only at the other instants.

Symbol	Meaning	SI unit
A	amplitude of the strain rate	s^{-1}
Ω	angular frequency	s^{-1}
R	radius of the initial sphere	m
(x_i, y_i, z_i)	initial position of a particle	m
$s(t) = (A/\Omega)\sin\Omega t$	dimensionless time function, used below	1

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, origin at the centre of the field.
- Material (Lagrangian) variables: the initial position (x_i, y_i, z_i) of a particle. The trajectories give the map $(x_i, y_i, z_i) \mapsto (x, y, z)$ at time t, and its inverse is used in part 3.
- Parametrisation of the initial sphere (Notes, footnote to Eq. 2.19): $x_i = -R/2 + R\sin\alpha\cos\beta$, $y_i = R\sin\alpha\sin\beta$, $z_i = R\cos\alpha$, with $0 \le \alpha \le \pi$ and $0 \le \beta < 2\pi$. This is a change of variable from the two surface labels (α, β) to the three coordinates, eliminated at the end.

PHASE 3

Step-by-Step Derivation

3.1 Velocity field, curl and divergence

Why this tool: $\mathbf{v} = \nabla\varphi$ (Eq. 2.34). Let $c \equiv \cos\Omega t$.

$$v_x = \frac{\partial\varphi}{\partial x} = 4Acx, \qquad v_y = \frac{\partial\varphi}{\partial y} = -2Acy, \qquad v_z = \frac{\partial\varphi}{\partial z} = -2Acz,$$

$$\boxed{\mathbf{v} = Ac\,(4x,\ -2y,\ -2z).}$$

Curl, component by component:

$$(\nabla\wedge\mathbf{v})_x=\partial_y v_z-\partial_z v_y=0-0,\quad (\nabla\wedge\mathbf{v})_y=\partial_z v_x-\partial_x v_z=0-0,\quad (\nabla\wedge\mathbf{v})_z=\partial_x v_y-\partial_y v_x=0-0,$$

so $\nabla\wedge\mathbf{v}=0$ (as for any gradient). Divergence:

$$\nabla\cdot\mathbf{v}=\partial_x(4Acx)+\partial_y(-2Acy)+\partial_z(-2Acz)=Ac\,(4-2-2)=0.$$

The potential satisfies $\nabla^2\varphi=Ac\,(4-2-2)=0$: it is harmonic, the flow is irrotational and incompressible for all t. The extension along x is balanced by the compression along y and z.

3.2 Streamlines, trajectories and path lines

Why this tool: streamlines come from the tangency condition at a frozen instant (Eq. 2.21), trajectories from the initial-value problem $d\mathbf{x}/dt=\mathbf{v}$ (Eq. 2.12), and path lines from eliminating t (Eq. 2.15).

Streamlines (at an instant with $c\neq 0$ the common factor Ac cancels):

$$\frac{dx}{4x}=\frac{dy}{-2y}=\frac{dz}{-2z}.$$

From the last equality, $dy/y=dz/z\Rightarrow \ln|y/y_0|=\ln|z/z_0|\Rightarrow y/z=y_0/z_0$. From the first equality, $dx/x=-2\,dy/y\Rightarrow \ln|x/x_0|=-2\ln|y/y_0|\Rightarrow$

$$\boxed{x\,y^2=x_0\,y_0^2,\qquad \frac{y}{z}=\frac{y_0}{z_0},}$$

the intersection of two surfaces (Eq. 2.15 type), independent of t.

Trajectories, with $\mathbf{x}(0)=(x_i,y_i,z_i)$. Each component decouples. Using Barrow's rule and $\int_0^t\cos\Omega t'\,dt'=\sin(\Omega t)/\Omega$:

$$\frac{dx}{dt} = 4A\cos(\Omega t)\,x \;\Rightarrow\; \int_{x_i}^{x}\frac{dx'}{x'} = 4A\int_0^t \cos\Omega t'\,dt' \;\Rightarrow\; \ln\frac{x}{x_i} = \frac{4A}{\Omega}\sin\Omega t,$$

$$\frac{dy}{dt} = -2A\cos(\Omega t)\,y \;\Rightarrow\; \ln\frac{y}{y_i} = -\frac{2A}{\Omega}\sin\Omega t, \qquad \frac{dz}{dt} = -2A\cos(\Omega t)\,z \;\Rightarrow\; \ln\frac{z}{z_i} = -\frac{2A}{\Omega}\sin\Omega t.$$

With $s(t) = (A/\Omega)\sin\Omega t$:

$$\boxed{x = x_i\,e^{4s(t)}, \qquad y = y_i\,e^{-2s(t)}, \qquad z = z_i\,e^{-2s(t)}.}$$

Path lines: eliminate s (that is, t). From the last two, $y/z = y_i/z_i$. Raise $y/y_i = e^{-2s}$ to the power -2: $(y/y_i)^{-2} = e^{4s} = x/x_i$, hence $xy^2 = x_i y_i^2$:

$$\boxed{x\,y^2 = x_i\,y_i^2, \qquad \frac{y}{z} = \frac{y_i}{z_i}.}$$

The path lines coincide with the streamlines of the flow (the direction field is steady; only the magnitude oscillates). The particles move back and forth along these curves: $s(t)$ oscillates between $\pm A/\Omega$, so x/x_i oscillates between $e^{-4A/\Omega}$ and $e^{4A/\Omega}$, and each particle returns to its initial position whenever $\sin\Omega t = 0$.

3.3 Fluid surface that is initially a sphere of radius R centred at $(-R/2, 0, 0)$

Why this tool: a fluid surface is the image under the trajectories of the initial surface (Eqs. 2.18-2.20); the surface labels (α, β) are then eliminated to obtain $f(x, y, z, t) = 0$.

Initial surface: $(x_i + R/2)^2 + y_i^2 + z_i^2 = R^2$, parametrised as in Phase 2. Trajectories map it to $x = x_i e^{4s}$, $y = y_i e^{-2s}$, $z = z_i e^{-2s}$, i.e. in terms of the labels,

$$\frac{x}{-R/2 + R\sin\alpha\cos\beta} = e^{4s}, \qquad \frac{y}{R\sin\alpha\sin\beta} = e^{-2s}, \qquad \frac{z}{R\cos\alpha} = e^{-2s}.$$

Eliminate (α, β) by inverting the map, $x_i = x\,e^{-4s}$, $y_i = y\,e^{2s}$, $z_i = z\,e^{2s}$, and substituting in the equation of the initial sphere:

$$\boxed{\left(x\,e^{-4s(t)} + \frac{R}{2}\right)^2 + \left(y\,e^{2s(t)}\right)^2 + \left(z\,e^{2s(t)}\right)^2 = R^2, \qquad s(t) = \frac{A}{\Omega}\sin\Omega t.}$$

This is an ellipsoid of revolution about the x-axis: its centre is at $x_c = -(R/2)\,e^{4s}$, its semi-axis along x is $R\,e^{4s}$ and its semi-axes along y and z are $R\,e^{-2s}$. The enclosed volume is $\frac{4\pi}{3}\,Re^{4s}\,R^2e^{-4s} = \frac{4\pi}{3}R^3$, constant, as required for an incompressible flow ($\nabla\cdot\mathbf{v} = 0$). At every instant with $\sin\Omega t = 0$ the surface is again the initial sphere.

Reading of the official handwritten solution

The statement fixes the centre at $(-R/2, 0, 0)$. In K8.pdf (part 3), as legible in the scan, the parametrisation is written $x_i - R/2 = R\sin\alpha\cos\beta$ and the final boxed equation contains $\left(xe^{-4A\sin(\Omega t)/\Omega} - R/2\right)^2$, which corresponds to a sphere centred at $x = +R/2$. For the sphere of the statement the sign in front of $R/2$ must be $+$, as derived above. Numerical test: for $R = 1$ m, $A = 0.6$ s^{-1}, $\Omega = 1.9$ s^{-1}, $t = 1.1$ s, a particle starting on the sphere of the statement ($\alpha = 0.8$, $\beta = 1.1$) satisfies the equation with $+R/2$ to 10^{-16}, whereas the $-R/2$ form leaves a residual of 0.349. The remaining handwritten results (components, streamlines, trajectories, paths) agree with the above.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: a pulsating pure-strain (stagnation-type) flow: the material is stretched along x and compressed in the (y, z) plane during the half cycles where $\cos\Omega t > 0$, and the opposite sense when $\cos\Omega t < 0$. The net displacement over a full period is zero.
- Limits: $A \to 0$ gives $s \to 0$ and the surface stays the initial sphere; $\Omega \to 0$ with $c \to 1$ gives the steady flow $\mathbf{v} = A(4x, -2y, -2z)$, for which $s \to At$ and the sphere is deformed without bound; at $\sin\Omega t = 0$ every trajectory returns to its starting point.
- Consistency: path lines equal streamlines for this flow, since the unit vector $\mathbf{v}/|\mathbf{v}|$ is time-independent; the volume of the fluid surface is conserved ($\nabla\cdot\mathbf{v} = 0$).
- Dimensions (SI): $[v] = [Ax] = \mathrm{s}^{-1}\mathrm{m} = \mathrm{m/s}$; $[s] = [A/\Omega] = 1$, so $e^{\pm ks}$ are dimensionless; the factors e^{4s}, e^{-2s} multiply lengths; in the ellipsoid equation each term has units m^2 ($x\,e^{-4s} + R/2$ is a length).

- Numerical cross-check: RK4 integration of the particle ODE with $A = 0.6$ s^{-1}, $\Omega = 1.9$ s^{-1} from $(0.5, 0.7, -0.4)$ m up to $t = 1.1$ s gives $(1.497122, 0.404533, -0.231162)$ m, the same as $x_i e^{4s}$, $y_i e^{-2s}$, $z_i e^{-2s}$ (agreement better than 10^{-11} m). Symbolic differentiation gives $\nabla \wedge \mathbf{v} = 0$ and $\nabla \cdot \mathbf{v} = 0$.

PROBLEM K9

Dipole | Polar coordinates | Acceleration | Fluid line

Oscillating Polar Flow (Planar Dipole)

Original statement · K9.pdf

OFFICIAL PROBLEM STATEMENT

For the planar velocity field

$$v_r = -A\sin(\Omega t)\,\frac{\sin\theta}{r^2}, \qquad v_\theta = A\sin(\Omega t)\,\frac{\cos\theta}{r^2},$$

where A and Ω are known constants,

1. Determine the vorticity.
2. Compute the expansion rate.
3. Comment on the existence of velocity potential and stream function and, if they do exist, compute them.
4. Find the acceleration for points along the line $\theta = 0$.
5. Obtain the trajectories and streamlines.
6. Determine the fluid line that at $t = 0$ is given by $\theta = 0$ and $0 < r < \infty$.

Source: K9.pdf (page 1). Notes.pdf, Chapter 2: curl and divergence in orthogonal coordinates Eqs. (2.5)-(2.6); potential Eq. (2.34); acceleration Eqs. (2.25)-(2.26); stream function Eqs. (2.40)-(2.45); trajectories Eq. (2.12); fluid lines Eqs. (2.16)-(2.17); stream lines Eq. (2.21).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar, unsteady flow in the plane (r, θ), for $r > 0$; the origin is singular and excluded. Write $\sigma(t) \equiv \sin \Omega t$; the velocity is the product of the time function σ and a steady spatial field. Both components decay as r^{-2}.
- Degrees of freedom: (r, θ) and time.
- Units: $v = A/r^2$ has units m/s, so A has units m^3/s.
- Hypothesis used in part 6: $A > 0, \Omega > 0$, and the time interval is such that the particle does not reach the singular point (discussed in Phase 4).

Symbol	Meaning	SI unit
A	dipole strength (per unit depth)	m^3/s
Ω	angular frequency	s^{-1}
(r_0, θ_0)	initial position of a particle	m, rad
λ	material label, $r_0 = \lambda$ on the line $\theta_0 = 0$	m

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame with the singular point at the origin; polar coordinates (r, θ), scale factors $(1, r)$, $x = r\cos\theta$, $y = r\sin\theta$, $\mathbf{e}_z = \mathbf{e}_r \wedge \mathbf{e}_\theta$.
- Operators (Eqs. 2.3, 2.5, 2.6): $\nabla\varphi = \partial_r\varphi\,\mathbf{e}_r + r^{-1}\partial_\theta\varphi\,\mathbf{e}_\theta$, $\nabla\cdot\mathbf{v} = r^{-1}\left[\partial_r(rv_r) + \partial_\theta v_\theta\right]$, $\omega_z = r^{-1}\left[\partial_r(rv_\theta) - \partial_\theta v_r\right]$.
- Stream function in polar coordinates. In Cartesian variables $v_x = \partial_y\psi$, $v_y = -\partial_x\psi$ (Eq. 2.41). Projecting on the polar basis, $v_r = v_x\cos\theta + v_y\sin\theta = \cos\theta\,\partial_y\psi - \sin\theta\,\partial_x\psi = r^{-1}\partial_\theta\psi$ and $v_\theta = -v_x\sin\theta + v_y\cos\theta = -\left(\sin\theta\,\partial_y\psi + \cos\theta\,\partial_x\psi\right) = -\partial_r\psi$, using $\partial_\theta = -r\sin\theta\,\partial_x + r\cos\theta\,\partial_y$ and $\partial_r = \cos\theta\,\partial_x + \sin\theta\,\partial_y$. Hence $v_r = r^{-1}\partial_\theta\psi$, $v_\theta = -\partial_r\psi$.
- For the acceleration, the coordinate-free form of Eq. (2.26) is used, since the Cartesian component form of Eq. (2.27) does not hold in polar coordinates.
- Material variables: initial position (r_0, θ_0) at $t = 0$.

PHASE 3

Step-by-Step Derivation

3.1 Vorticity

Why this tool: $\omega_z = r^{-1}\left[\partial_r(rv_\theta) - \partial_\theta v_r\right]$ is twice the angular velocity of the fluid element (Eq. 2.52).

$$r\,v_\theta = \frac{A\sigma\cos\theta}{r} \Rightarrow \frac{\partial(rv_\theta)}{\partial r} = -\frac{A\sigma\cos\theta}{r^2}, \qquad \frac{\partial v_r}{\partial\theta} = -\frac{A\sigma\cos\theta}{r^2},$$

$$\omega_z = \frac{1}{r}\left[-\frac{A\sigma\cos\theta}{r^2} + \frac{A\sigma\cos\theta}{r^2}\right] = \boxed{0.}$$

3.2 Expansion rate

Why this tool: $\nabla \cdot \mathbf{v}$ is the rate of volume change per unit volume (Eqs. 2.62-2.63).

$$r\, v_r = -\frac{A\sigma \sin\theta}{r} \Rightarrow \frac{\partial (r v_r)}{\partial r} = \frac{A\sigma \sin\theta}{r^2}, \qquad \frac{\partial v_\theta}{\partial \theta} = -\frac{A\sigma \sin\theta}{r^2},$$

$$\nabla \cdot \mathbf{v} = \frac{1}{r}\left[\frac{A\sigma \sin\theta}{r^2} - \frac{A\sigma \sin\theta}{r^2}\right] = \boxed{0.}$$

3.3 Existence of potential and stream function

Because $\nabla \wedge \mathbf{v} = 0$ (3.1) in the region $r > 0$, a velocity potential exists (Eq. 2.34): $\mathbf{v} = \nabla\varphi$. Because the planar flow is also solenoidal (3.2), a stream function exists (Eq. 2.41). Both are single-valued here: the circulation around the origin is $\oint v_\theta\, r\, d\theta = A\sigma \int_0^{2\pi} \cos\theta\, d\theta / r = 0$ and the flux through a circle $\oint v_r\, r\, d\theta = -A\sigma \int_0^{2\pi} \sin\theta\, d\theta / r = 0$.

Potential. From $v_r = \partial_r \varphi = -A\sigma \sin\theta / r^2$, integrate in r (primitive of r^{-2} is $-r^{-1}$): $\varphi = A\sigma \sin\theta / r + f(\theta, t)$. Then

$$v_\theta = \frac{1}{r}\frac{\partial \varphi}{\partial \theta} = \frac{A\sigma \cos\theta}{r^2} + \frac{1}{r}\frac{\partial f}{\partial \theta} = \frac{A\sigma \cos\theta}{r^2} \implies \frac{\partial f}{\partial \theta} = 0,$$

$$\boxed{\varphi = \frac{A \sin(\Omega t) \sin\theta}{r} + \varphi_0(t) = A \sin(\Omega t)\, \frac{y}{x^2 + y^2} + \varphi_0(t).}$$

Stream function. From $r\, v_r = \partial_\theta \psi = -A\sigma \sin\theta / r$, integrate in θ: $\psi = A\sigma \cos\theta / r + g(r, t)$. Then

$$v_\theta = -\frac{\partial \psi}{\partial r} = \frac{A\sigma \cos\theta}{r^2} - \frac{\partial g}{\partial r} = \frac{A\sigma \cos\theta}{r^2} \implies \frac{\partial g}{\partial r} = 0,$$

$$\boxed{\psi = \frac{A\sin(\Omega t)\cos\theta}{r} + \psi_0(t) = A\sin(\Omega t)\,\frac{x}{x^2+y^2} + \psi_0(t).}$$

This is the field of a planar dipole aligned with y whose strength oscillates as $\sin\Omega t$.

3.4 Acceleration along $\theta = 0$

Why this tool: $\mathbf{a} = D\mathbf{v}/Dt$ (Eq. 2.25). Since $\nabla \wedge \mathbf{v} = 0$, Eq. (2.26) reduces to $\mathbf{a} = \partial_t \mathbf{v} + \nabla(|\mathbf{v}|^2/2)$ in any coordinate system.

$|\mathbf{v}|^2 = v_r^2 + v_\theta^2 = A^2\sigma^2\left(\sin^2\theta + \cos^2\theta\right)/r^4 = A^2\sigma^2/r^4$, so

$$\frac{|\mathbf{v}|^2}{2} = \frac{A^2\sigma^2}{2r^4}, \qquad \frac{\partial}{\partial r}\left(\frac{|\mathbf{v}|^2}{2}\right) = -\frac{2A^2\sigma^2}{r^5}, \qquad \frac{1}{r}\frac{\partial}{\partial\theta}\left(\frac{|\mathbf{v}|^2}{2}\right) = 0.$$

Local term, with $\partial_t\sigma = \Omega\cos\Omega t$: $\partial_t v_r = -A\Omega\cos(\Omega t)\sin\theta/r^2$ and $\partial_t v_\theta = A\Omega\cos(\Omega t)\cos\theta/r^2$. Therefore, for any θ,

$$a_r = -\frac{A\Omega\cos(\Omega t)\sin\theta}{r^2} - \frac{2A^2\sin^2(\Omega t)}{r^5}, \qquad a_\theta = \frac{A\Omega\cos(\Omega t)\cos\theta}{r^2}.$$

On the line $\theta = 0$ ($\sin\theta = 0$, $\cos\theta = 1$):

$$\boxed{a_r = -\frac{2A^2\sin^2(\Omega t)}{r^5}, \qquad a_\theta = \frac{A\Omega\cos(\Omega t)}{r^2}.}$$

3.5 Trajectories and streamlines

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12) and streamlines the tangency condition (Eq. 2.21).

Streamlines at a frozen instant ($\sigma \neq 0$): $dr/v_r = r\,d\theta/v_\theta$ gives

$$\frac{dr}{r\,d\theta} = \frac{v_r}{v_\theta} = -\frac{\sin\theta}{\cos\theta} \implies \frac{dr}{r} = -\tan\theta\,d\theta \implies \ln\frac{r}{r_0} = \ln\left|\frac{\cos\theta}{\cos\theta_0}\right|, \qquad \boxed{r = r_0\frac{\cos\theta}{\cos\theta_0} = C\cos\theta,}$$

because $\int -\tan\theta\,d\theta = \ln|\cos\theta|$. In Cartesian variables $x^2 + y^2 = Cx$: circles of diameter C centred at $(C/2, 0)$, tangent to the y-axis at the origin. This is the dipole pattern, independent of t.

Trajectories: the system $dr/dt = v_r$, $r\,d\theta/dt = v_\theta$ contains the same ratio, so dividing the equations removes t and $\sigma(t)$ and gives the same relation; therefore the path lines are the same circles, $r = r_0\cos\theta/\cos\theta_0$.

Time law. Insert $r^3 = r_0^3\cos^3\theta/\cos^3\theta_0$ in $d\theta/dt = v_\theta/r = A\sigma\cos\theta/r^3$:

$$\frac{d\theta}{dt} = \frac{A\cos^3\theta_0}{r_0^3}\,\frac{\sin\Omega t}{\cos^2\theta} \implies \cos^2\theta\,d\theta = \frac{A\cos^3\theta_0}{r_0^3}\sin\Omega t\,dt.$$

Integrate with Barrow's rule between $(\theta_0, 0)$ and (θ, t). For the left side use $\cos^2\theta' = (1 + \cos 2\theta')/2$, whose primitive is $\theta'/2 + \sin 2\theta'/4$; for the right side $\int_0^t \sin\Omega t'\,dt' = (1 - \cos\Omega t)/\Omega$:

$$\frac{\theta - \theta_0}{2} + \frac{\sin 2\theta - \sin 2\theta_0}{4} = \frac{A\cos^3\theta_0}{r_0^3\Omega}\left(1 - \cos\Omega t\right),$$

$$\boxed{\theta - \theta_0 + \frac{\sin 2\theta - \sin 2\theta_0}{2} = \frac{2A\cos^3\theta_0}{r_0^3\,\Omega}\left(1 - \cos\Omega t\right), \qquad r = r_0\frac{\cos\theta}{\cos\theta_0}.}$$

The right-hand side vanishes at $\Omega t = 2n\pi$, so the particle oscillates with the period $2\pi/\Omega$ and returns to its initial position at the end of each period.

3.6 Fluid line initially at $\theta = 0$, $0 < r < \infty$

Why this tool: the fluid line is the image of the initial curve under the trajectories (Eq. 2.17), with the material label eliminated afterwards. Label the particles by $\lambda = r_0$, with $\theta_0 = 0$ (so $\cos\theta_0 = 1, \sin 2\theta_0 = 0$).

From 3.5, the path and the time law reduce to

$$\frac{r}{\cos\theta} = \lambda, \qquad \theta + \frac{\sin 2\theta}{2} = \frac{2A}{\lambda^3 \Omega}\,(1 - \cos\Omega t)\,.$$

Eliminate λ (change of variable from label to current position) with $\lambda = r/\cos\theta$, $\lambda^{-3} = \cos^3\theta / r^3$:

$$\theta + \frac{\sin 2\theta}{2} = \frac{2A\cos^3\theta}{r^3\,\Omega}\,(1 - \cos\Omega t) \implies \boxed{\frac{r^3}{2\cos^3\theta}\left(\theta + \frac{\sin 2\theta}{2}\right) = \frac{A}{\Omega}\,[1 - \cos(\Omega t)]\,.}$$

At $t = 0$ the right-hand side is 0 and the equation gives $\theta = 0$ (the initial line), as it must.

Reading of the official handwritten solution

In K9.pdf (part 6) the boxed final equation reads, as legible in the scan, $\frac{r^3}{2\cos^3\theta}\left(\theta + \frac{\sin 2\theta}{2}\right) = \frac{A}{\Omega}\,[1 + \cos(\Omega t)]$, whereas the line immediately above it in the same solution has $\frac{A}{\Omega}\,[1 - \cos(\Omega t)] = \frac{\lambda^3}{2}\left(\theta + \frac{\sin 2\theta}{2}\right)$. With $1 + \cos\Omega t$ the right-hand side would be $2A/\Omega$ at $t = 0$ and the initial line $\theta = 0$ would not be recovered; the correct sign is $1 - \cos\Omega t$, as derived above. Numerical test with $A = 0.7\ \mathrm{m^3/s}$, $\Omega = 1.6\ \mathrm{s^{-1}}$, $\lambda = 1.2\ \mathrm{m}$, $t = 1.4\ \mathrm{s}$: integrating the particle ODE gives $r^3(\theta + \sin 2\theta/2)/(2\cos^3\theta) = 0.708908$, equal to $(A/\Omega)(1 - \cos\Omega t) = 0.708908$, while the "plus" form would give 0.166092.

PHASE 4

Interpretation, Limits and Dimensional Check

- The field is the oscillating field of a planar dipole: irrotational and incompressible, with circular streamlines through the singular point; the particles slosh back and forth along the circles (θ varies with $1 - \cos\Omega t \geq 0$, so for $A > 0$ the particle first moves towards

larger θ and then returns).

- Existence of the fluid line: since $\theta + \frac{1}{2}\sin 2\theta \le \pi/2$ for $\theta \le \pi/2$, particles of the line with $\dfrac{2A}{\lambda^3\Omega}\cdot 2 \ge \dfrac{\pi}{2}$, i.e. $\lambda^3 \le \dfrac{8A}{\pi\Omega}$, reach $\theta = \pi/2$ and $r = 0$ (the singular point) within a cycle. The formula holds for the particles that stay at $|\theta| < \pi/2$ ($\lambda > (8A/\pi\Omega)^{1/3}$, which is 1.0367 m for $A = 0.7$ m^3/s and $\Omega = 1.6$ s^{-1}).
- Consistency: parts 1-2 imply 3 (potential and stream function exist); $\psi = \text{const}$ reproduces the streamlines $\cos\theta/r = \text{const}$ of 3.5; the acceleration of 3.4 is the field value at $\theta = 0$ (the acceleration of the particle that is there at time t).
- Dimensions (SI): $[v] = [A/r^2] = \mathrm{m/s}$; $[\varphi] = [\psi] = [A/r] = \mathrm{m^2/s}$; $[A\Omega/r^2] = (\mathrm{m^3/s})(\mathrm{s^{-1}})/\mathrm{m^2} = \mathrm{m/s^2}$ and $[A^2/r^5] = (\mathrm{m^6/s^2})/\mathrm{m^5} = \mathrm{m/s^2}$; $A/(r_0^3\Omega)$ is dimensionless; the fluid-line equation has units $\mathrm{m^3}$ on both sides.
- Numerical cross-check: symbolic differentiation returns $\omega_z = 0$, $\nabla\cdot\mathbf{v} = 0$, $\nabla\varphi = \mathbf{v}$ and the polar stream-function relations exactly, and the full material derivative in polar components gives $a_r = -2A^2\sigma^2/r^5$, $a_\theta = A\Omega\cos\Omega t/r^2$ at $\theta = 0$. For the time law of 3.5, with $A = 0.7$ m^3/s, $\Omega = 1.6$ s^{-1}, $r_0 = 1.5$ m, $\theta_0 = 0.4$ rad, $t = 0.8$ s, the ODE integration gives 0.144498 for the left side and the formula 0.144498 for the right side.

PROBLEM K10

Unsteady | Potential | Stream function | Pure shear

Planar Flow with Exponentially Varying Shear (Quiz, 30 September 2013)

Original statement · K10.pdf

OFFICIAL PROBLEM STATEMENT

Given the steady planar motion described by the velocity components

$$v_x = Ae^{\Omega t}\, y \qquad \text{and} \qquad v_y = Be^{\Omega t}\, x,$$

where A, B, and Ω are constants.

1. Determine the value of the volume dilation rate $\nabla \cdot \vec{v}$.
2. Obtain the value of B for which the resulting flow is irrotational. Adopt this particular value in what follows.
3. Calculate the stream line that intersects the point $(x, y) = (x_0, y_0)$ at a given instant of time t. Sketch the resulting stream lines.
4. Compute the trajectory of a fluid particle located initially at $(x, y) = (x_0, y_0)$.
5. If they exist, calculate the velocity potential φ and the stream function ψ.
6. Determine the rate-of-strain tensor $\bar{\bar{T}}_d$.
7. Describe the deformation of a fluid element of square shape whose sides of length dl are initially oriented parallel to the axes.

Source: K10.pdf (page 1). Notes.pdf, Chapter 2: expansion rate Eqs. (2.62)-(2.63); vorticity Eq. (2.52); stream lines Eq. (2.21); trajectories Eq. (2.12); potential Eq. (2.34); stream function Eqs. (2.40)-(2.45); velocity gradient and rate-of-strain tensor Eqs. (2.46)-(2.48); square element Eqs. (2.56)-(2.61).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar flow with a velocity that is linear in the coordinates and multiplied by the common time factor $e^{\Omega t}$. Two spatial degrees of freedom and an explicit time dependence.
- Although the statement calls the motion "steady", the velocity depends explicitly on t whenever $\Omega \neq 0$; the flow is unsteady in general (and part 3 asks for the streamline "at a given instant of time t"). The steady case is recovered for $\Omega = 0$. The general case $\Omega \neq 0$ is solved below, with $A \neq 0$.
- The constants A, B and Ω have units s^{-1} (so that $v = Ae^{\Omega t}y$ is a velocity).

Symbol	Meaning	SI unit
A, B	shear constants	s^{-1}
Ω	rate of the exponential time dependence	s^{-1}
(x_0, y_0)	position at the reference (initial) instant	m
$\tau(t) = (A/\Omega)\left(e^{\Omega t} - 1\right)$	dimensionless time variable defined in 3.4	1

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y)$, origin at the stagnation point $\mathbf{v} = 0$. Gradient convention $(\nabla \mathbf{v})_{ij} = \partial v_j / \partial x_i$.
- Change of time variable for part 4: $\tau(t) = \dfrac{A}{\Omega}\left(e^{\Omega t} - 1\right)$, with $d\tau = Ae^{\Omega t}dt$ and $\tau(0) = 0$. In terms of τ the unsteady trajectory system becomes autonomous with constant coefficients.
- Change of basis for the sum and difference coordinates $u = x + y$, $w = x - y$ (a rotation by 45° and a scaling by $\sqrt{2}$) used to decouple the trajectory equations; this is the same basis that diagonalises $\bar{\bar{T}}_d$ in part 6.

PHASE 3

Step-by-Step Derivation

3.1 Volume dilation rate

Why this tool: $\nabla \cdot \mathbf{v}$ is the rate at which a material volume changes per unit volume (Eqs. 2.62-2.63).

$$\nabla \cdot \mathbf{v} = \frac{\partial}{\partial x}\left(Ae^{\Omega t}y\right) + \frac{\partial}{\partial y}\left(Be^{\Omega t}x\right) = 0 + 0 = \boxed{0}$$

for any values of A, B and Ω: the flow is incompressible.

3.2 Condition for irrotational flow

Why this tool: the flow is irrotational when $\boldsymbol{\omega} = \nabla \wedge \mathbf{v} = 0$ (text after Eq. 2.33).

$$\omega_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = Be^{\Omega t} - Ae^{\Omega t} = (B - A)e^{\Omega t}.$$

Since $e^{\Omega t} > 0, \omega_z = 0$ for all t if and only if

$$\boxed{B = A.}$$

From now on $\mathbf{v} = Ae^{\Omega t}(y, x)$.

3.3 Streamlines at a given instant

Why this tool: streamlines are tangent to the velocity at a frozen instant (Eq. 2.21); t is a parameter.

$$\frac{dx}{v_x} = \frac{dy}{v_y} \implies \frac{dx}{Ae^{\Omega t}y} = \frac{dy}{Ae^{\Omega t}x} \implies \frac{dx}{y} = \frac{dy}{x} \implies x\,dx = y\,dy,$$

where the common non-zero factor $Ae^{\Omega t}$ cancels. Integrate between (x_0, y_0) and (x, y) (Barrow's rule):

$$\left[\frac{x'^2}{2}\right]_{x_0}^{x} = \left[\frac{y'^2}{2}\right]_{y_0}^{y} \implies \boxed{y^2 - x^2 = y_0^2 - x_0^2.}$$

These are rectangular hyperbolae with asymptotes $y = \pm x$; the shape does not depend on t (only the speed does). For $y_0^2 > x_0^2$ the branches open towards $\pm y$, for $y_0^2 < x_0^2$ towards $\pm x$, and $y_0^2 = x_0^2$ gives the straight lines $y = \pm x$. Sketch (in words), for $Ae^{\Omega t} > 0$: v_x has the sign of y and v_y the sign of x, so in the first and third quadrants the flow moves away from the origin along the diagonal $y = x$, and in the second and fourth quadrants towards the origin along $y = -x$. The origin is a stagnation point; it is a saddle, with $y = -x$ the incoming and $y = x$ the outgoing direction. For $Ae^{\Omega t} < 0$ all arrows reverse.

3.4 Trajectory of the particle starting at (x_0, y_0)

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}(\mathbf{x}, t)$ with $\mathbf{x}(0) = (x_0, y_0)$ (Eq. 2.12). Because $\mathbf{v}$ is a separable product of a function of time and a function of position, the change of time variable τ of Phase 2 makes the system autonomous. With $d\tau = Ae^{\Omega t}dt$:

$$\frac{dx}{d\tau} = y, \qquad \frac{dy}{d\tau} = x.$$

Sum and difference: $u = x + y$ and $w = x - y$ satisfy

$$\frac{du}{d\tau} = y + x = u, \qquad \frac{dw}{d\tau} = y - x = -w.$$

Integrate with Barrow's rule between $\tau = 0$ and τ: $\ln(u/u_0) = \tau$ and $\ln(w/w_0) = -\tau$, i.e. $u = (x_0 + y_0)e^{\tau}$ and $w = (x_0 - y_0)e^{-\tau}$. Return to $x = (u + w)/2$, $y = (u - w)/2$:

$$\boxed{x = x_0 \cosh\tau + y_0 \sinh\tau, \qquad y = y_0 \cosh\tau + x_0 \sinh\tau, \qquad \tau = \frac{A}{\Omega}\left(e^{\Omega t} - 1\right).}$$

Path lines: eliminate τ using $\cosh^2\tau - \sinh^2\tau = 1$:

$$y^2 - x^2 = (y_0^2 - x_0^2)\left(\cosh^2\tau - \sinh^2\tau\right) = y_0^2 - x_0^2,$$

the same hyperbolae as the streamlines. (The path lines coincide with the streamlines because the direction of $\mathbf{v}$ does not depend on time.)

Logarithmic form (equivalent, as in the handwritten solution). Along the path $y = \pm\sqrt{x^2 + y_0^2 - x_0^2}$, so $dx/dt = \pm A e^{\Omega t}\sqrt{x^2 + C}$ with $C = y_0^2 - x_0^2$. Since $\dfrac{d}{dx}\ln\left(x + \sqrt{x^2 + C}\right) = \dfrac{1}{\sqrt{x^2 + C}}$, Barrow's rule gives, for $y_0 > 0$ (upper sign),

$$\ln\frac{x + \sqrt{x^2 + y_0^2 - x_0^2}}{x_0 + y_0} = \frac{A}{\Omega}\left(e^{\Omega t} - 1\right),$$

which is $\ln[(x+y)/(x_0+y_0)] = \tau$ obtained from $u = u_0 e^{\tau}$. For $y_0 < 0$ the lower branch of the square root applies.

3.5 Velocity potential and stream function

Why this tool: the flow is irrotational (3.2) so a potential exists (Eq. 2.34); it is planar and solenoidal (3.1) so a stream function exists (Eq. 2.41).

Potential: $\partial_x\varphi = v_x = A e^{\Omega t} y$ gives $\varphi = A e^{\Omega t} xy + f(y,t)$; $\partial_y\varphi = A e^{\Omega t} x + f' = v_y = A e^{\Omega t} x$ gives $f' = 0$:

$$\boxed{\varphi = A e^{\Omega t}\, xy + \varphi_0(t).}$$

Stream function: $\partial_y\psi = v_x = A e^{\Omega t} y$ gives $\psi = \frac{1}{2} A e^{\Omega t} y^2 + g(x,t)$; $-\partial_x\psi = -g' = v_y = A e^{\Omega t} x$ gives $g = -\frac{1}{2} A e^{\Omega t} x^2$:

$$\boxed{\psi = \frac{A e^{\Omega t}}{2}\left(y^2 - x^2\right) + \psi_0(t).}$$

Check: $\partial_y\psi = A e^{\Omega t} y = v_x$ and $-\partial_x\psi = A e^{\Omega t} x = v_y$. The level curves $\psi = \text{const}$ are the hyperbolae of 3.3 (Eq. 2.42), and are orthogonal to the level curves of φ ($xy = \text{const}$).

Reading of the official handwritten solution

K10.pdf (part 5) gives $\psi = Ae^{\Omega t}(y^2 - x^2) + \psi_0$, without the factor $1/2$. That expression would give $\partial_y \psi = 2Ae^{\Omega t} y$, twice the velocity component v_x. The relations written just above that line, $\partial_y \psi = Ae^{\Omega t} y$ and $\partial_x \psi = -Ae^{\Omega t} x$, are correct, and integrating them produces the factor $1/2$. The correct result is $\psi = \frac{1}{2} Ae^{\Omega t}(y^2 - x^2) + \psi_0$. The statement also calls the motion "steady", although the velocity depends on t through $e^{\Omega t}$ (see Phase 1). The velocity potential $\varphi = Ae^{\Omega t} xy + \varphi_0$ in the handwritten solution is correct.

3.6 Rate-of-strain tensor

Why this tool: $\bar{\bar{T}}_d = \frac{1}{2}\left[\nabla \mathbf{v} + (\nabla \mathbf{v})^T\right]$ (Eq. 2.47) gives the pure strain of the material elements.

$$\nabla \mathbf{v} = \begin{pmatrix} \partial_x v_x & \partial_x v_y \\ \partial_y v_x & \partial_y v_y \end{pmatrix} = Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \boxed{\bar{\bar{T}}_d = Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \nabla \mathbf{v}.}$$

It is symmetric, so $\bar{\bar{T}}_r = \frac{1}{2}\left[\nabla \mathbf{v} - (\nabla \mathbf{v})^T\right] = 0$, consistent with $\omega_z = 0$. Its eigenvalues follow from $\lambda^2 - (Ae^{\Omega t})^2 = 0$: $\lambda = \pm Ae^{\Omega t}$, with principal directions $(1, 1)/\sqrt{2}$ (extension, for $Ae^{\Omega t} > 0$) and $(1, -1)/\sqrt{2}$ (compression); these are the outgoing and incoming diagonals found in 3.3. The trace is zero ($\nabla \cdot \mathbf{v} = 0$).

3.7 Deformation of a square fluid element

Why this tool: a material segment evolves as $d\mathbf{l}(t + dt) = d\mathbf{l} + (d\mathbf{l} \cdot \nabla \mathbf{v})\, dt$ (Eqs. 2.46 and 2.56-2.57). With the sides along the axes:

$$dl\,(1,0) \;\to\; dl\,(1,0) + dl\,(1,0)\, Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} dt = dl\,\left(1,\; Ae^{\Omega t} dt\right),$$

$$dl\,(0,1) \;\to\; dl\,(0,1) + dl\,(0,1)\, Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} dt = dl\,\left(Ae^{\Omega t} dt,\; 1\right).$$

The horizontal side tilts upward by the angle $Ae^{\Omega t} dt$ and the vertical side tilts towards $+x$ by the same angle, so the two sides rotate towards each other, and their (initially right) angle decreases by $2Ae^{\Omega t} dt$. There is no net rotation ($\bar{\bar{T}}_r = 0$): the square deforms into a

rhombus-like parallelogram by pure shear strain, with strain rate $\dot{\gamma} = \partial_x v_y + \partial_y v_x = 2Ae^{\Omega t}$ (the rate at which the angle between the sides decreases, Notes after Eq. 2.61). The area changes only at second order ($dl^2\,\left[1 - (Ae^{\Omega t}dt)^2\right]$), as required by $\nabla \cdot \mathbf{v} = 0$. Equivalently, the diagonal along $y = x$ is stretched at the rate $Ae^{\Omega t}$ and the diagonal along $y = -x$ is compressed at the same rate.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: it is the stagnation-point flow of K7 rotated by 45° (principal axes along the diagonals, no rotation), with a strain rate $Ae^{\Omega t}$ that grows or decays exponentially in time.
- Steady limit $\Omega \to 0$: $\tau \to At$, and the trajectories become $x = x_0 \cosh At + y_0 \sinh At$, $y = y_0 \cosh At + x_0 \sinh At$, the standard stagnation-point trajectories. For $\Omega < 0$ the time variable saturates at $\tau_\infty = A/|\Omega|$, so particles approach finite limit positions (the strain rate dies out) instead of escaping.
- Particles on the incoming diagonal $y_0 = -x_0$ satisfy $u = 0$ for all t and $w = w_0 e^{-\tau}$: they move towards the origin as τ grows, but reach it only as $\tau \to \infty$ (for $\Omega > 0$, $A > 0$, as $t \to \infty$).
- Consistency: $\bar{\bar{T}}_d$ symmetric traceless, $\omega_z = 0$, $\nabla \cdot \mathbf{v} = 0$; $\psi =$ const reproduces the streamlines of 3.3 and the paths of 3.4.
- Dimensions (SI): $[Ae^{\Omega t}y] = \mathrm{s}^{-1}\mathrm{m} = \mathrm{m/s}$; $[\tau] = [A/\Omega] = 1$; $[\varphi] = [Axy] = \mathrm{m}^2/\mathrm{s}$; $[\psi] = [Ay^2] = \mathrm{m}^2/\mathrm{s}$; $[\bar{\bar{T}}_d] = [A] = \mathrm{s}^{-1}$; $[\dot{\gamma}] = \mathrm{s}^{-1}$.
- Numerical cross-check: RK4 integration of the particle ODE with $A = B = 0.8\ \mathrm{s}^{-1}$, $\Omega = 0.9\ \mathrm{s}^{-1}$, $(x_0, y_0) = (0.5, 1.2)\ \mathrm{m}$ up to $t = 1.7\ \mathrm{s}$ gives $(21.17827, 21.20635)\ \mathrm{m}$, equal to the cosh/sinh formulas ($\tau = 3.216157$), with $y^2 - x^2 = 1.19\ \mathrm{m}^2$ equal to $y_0^2 - x_0^2$, and $\ln[(x+y)/(x_0+y_0)] = 3.216157 = \tau$.